

DETAILED PROJECT REPORT (DPR)

LITHIUM ION BATTERY RECYCLING & HYDROMETALLURGY PLANT



**5 TONS
PER DAY
CAPACITY**

LITHIUM ION BATTERY
RECYCLING TILL
BLACK MASS



**0.5 TONS
PER DAY
CAPACITY**

HYDROMETALLURGY
PROCESS OF BLACK
MASS RECOVERED

CLIENT:

 **S. D. SALES**

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SAS NAGAR, PUNJAB - 140603

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SUSTAINABLE
BATTERY RECYCLING



RESOURCE
RECOVERY



CLEAN TECHNOLOGY
& GREEN PROCESS



HIGH VALUE
METAL RECOVERY



ENVIRONMENT
COMPLIANT

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1. Introduction

EPR Nexuss Solutions Private limited stands at the forefront of sustainable infrastructure development, bringing unmatched expertise in the planning, design, and execution of recycling plant setups. With a proven track record across diverse waste management and resource recovery projects, the organization has established itself as a trusted partner for delivering efficient, compliant, and future-ready recycling solutions.

Over the years, EPR Nexuss Solutions has successfully conceptualized and implemented a wide range of recycling facilities, tailored to varying waste streams and regional requirements. Our comprehensive approach spans feasibility assessment, regulatory approvals, engineering design, technology selection, and end-to-end project execution. This integrated capability ensures that every project is not only environmentally sound but also economically viable and operationally robust.

A key differentiator for us is its highly skilled research and technical team. Comprising experts from environmental engineering, material science, process optimization, and sustainability domains, the team continuously works on advancing recycling technologies and improving plant efficiencies. Their in-depth understanding of evolving regulatory frameworks, combined with hands-on industry experience, enables them to design systems that meet stringent compliance standards while maximizing resource recovery.

The organization's research-driven approach allows it to adopt cutting-edge methodologies, optimize process flows, and incorporate innovative solutions tailored to specific project needs. This ensures that clients benefit from state-of-the-art infrastructure that is scalable, efficient, and aligned with global best practices.

With a strong commitment to sustainability, innovation, and excellence, EPR Nexuss continues to lead the way in developing recycling ecosystems that contribute meaningfully to circular economy goals and environmental conservation.

2. Executive Summary

The proposed project involves the establishment of a Lithium-Ion Battery Recycling Plant with a processing capacity of 5 Tons per Day (TPD) for Black Mass production and 0.5 TPD for Hydrometallurgical processing. The facility is designed to recover valuable metals such as Lithium, Graphite, Cobalt, Nickel, and Manganese from end-of-life lithium-ion batteries, supporting sustainable resource utilization and circular economy objectives.

With the rapid expansion of electric vehicles, renewable energy storage systems, and electronic devices, the generation of lithium-ion battery waste is increasing significantly. This project aims to address this growing challenge by deploying advanced and environmentally compliant recycling technologies to efficiently process battery waste and recover high-value materials.

Project Configuration

The proposed plant will consist of two primary operational units:

- Mechanical Recycling Unit (5 TPD): Responsible for safe discharge, dismantling, shredding, and segregation of batteries to produce black mass.
- Hydrometallurgical Processing Unit (0.5 TPD): Designed for chemical treatment and refining of black mass to extract and purify critical metals.

Project Development Scope

The client will undertake land acquisition and civil construction, including development of plant infrastructure, utilities, storage areas, and supporting facilities. The project report outlines the technical framework, process flow, equipment requirements, and operational strategy for efficient plant functioning.

Market Opportunity

The lithium-ion battery recycling sector is witnessing strong growth due to:

- Increasing adoption of electric mobility
- Regulatory push under Battery Waste Management Rules, 2022
- Rising demand for critical metals in battery manufacturing
- Need for environmentally responsible waste management solutions

This project is well-positioned to cater to both compliance-driven demand and industrial raw material recovery requirements.

Operational and Environmental Considerations

The plant will be designed in accordance with applicable environmental and safety regulations, including:

- Central Pollution Control Board (CPCB) guidelines

- Hazardous Waste Management Rules
- Battery Waste Management Rules, 2022

Adequate provisions for effluent treatment, air pollution control, and safe handling of hazardous materials will be integrated into the plant design to ensure environmentally sound operations.

Strategic Significance

- Supports circular economy and resource recovery initiatives
- Reduces dependency on imported critical minerals
- Aligns with national sustainability and compliance frameworks
- Scalable model to accommodate future capacity expansion

Conclusion

The proposed Lithium-Ion Battery Recycling Plant represents a forward-looking and environmentally responsible initiative. By integrating mechanical and hydrometallurgical processes, the project aims to establish a robust and efficient system for recovery of valuable materials from battery waste, contributing to both economic and environmental sustainability.

3. Business Overview

The proposed project focuses on establishing a **Lithium-Ion Battery Recycling Plant** with an integrated system for **black mass production (5 TPD)** and **hydrometallurgical processing (0.5 TPD)**. The plant is aimed at recovering critical and high-value metals from end-of-life lithium-ion batteries, thereby supporting the growing demand for sustainable raw materials in the battery and energy storage industry.

The business operates at the intersection of **waste management, resource recovery, and environmental compliance**, addressing the increasing volume of battery waste generated from electric vehicles, consumer electronics, and renewable energy storage systems. With regulatory frameworks such as the **Battery Waste Management Rules, 2022**, mandating proper recycling and recovery, the demand for authorized recycling facilities is steadily increasing.

Nature of Business

The plant will function as an **authorized recycler of lithium-ion batteries**, undertaking:

- Collection and procurement of end-of-life batteries and scrap
- Safe handling, dismantling, and mechanical processing to generate black mass
- Chemical processing of black mass to recover valuable metals
- Supplying recovered materials back to battery manufacturers and allied industries

Core Business Activities

- **Battery Recycling & Processing:** End-to-end recycling from battery discharge to material recovery
- **Material Recovery & Refining:** Extraction of Lithium, Cobalt, Nickel, and other metals
- **Waste Management Services:** Supporting producers in meeting EPR obligations
- **Resource Supply Chain Integration:** Supplying recovered materials to manufacturers

Target Market

The plant will cater to a diverse and growing customer base, including:

- **Battery manufacturers and gigafactories**
- **Electric vehicle (EV) manufacturers and OEMs**
- **E-waste aggregators and dismantlers**
- **Producers and importers requiring EPR compliance**
- **Metal refining and chemical industries**

Revenue Streams

The business model is driven by multiple revenue channels:

- Sale of recovered metals and processed materials
- Recycling and processing charges (tolling model)

- EPR compliance partnerships with producers and importers
- Long-term supply agreements with battery manufacturers

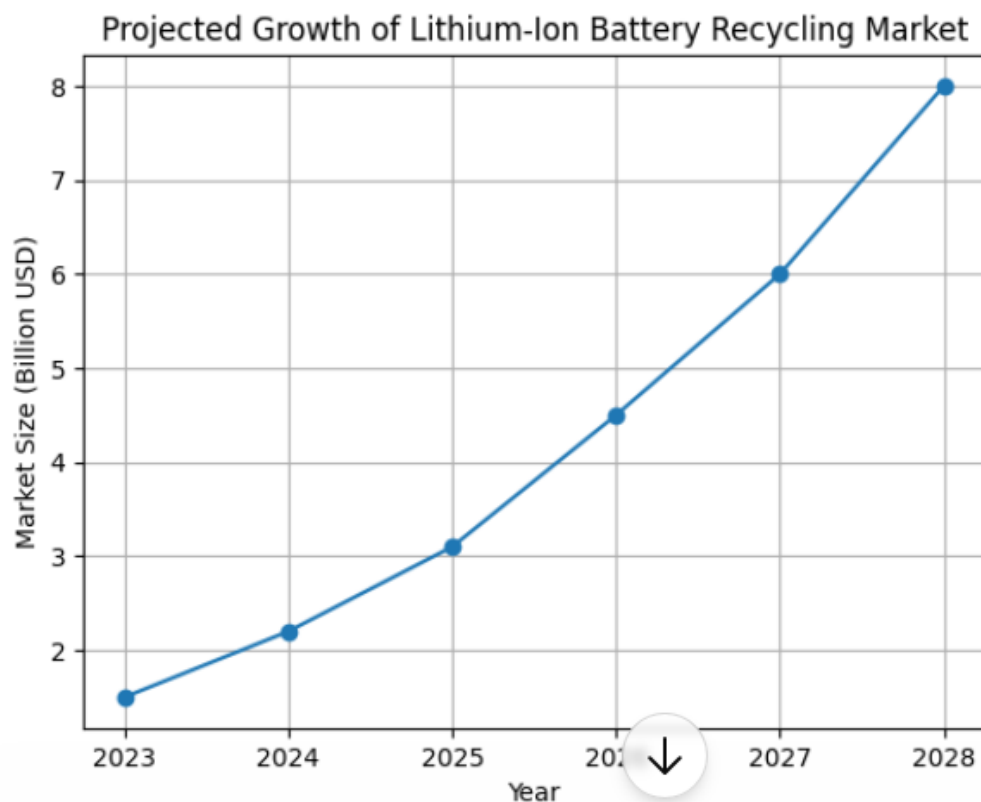
Market Positioning

The proposed facility is positioned as a **technology-driven and compliance-oriented recycling unit**, focusing on:

- High recovery efficiency and material purity
- Environmentally compliant operations
- Reliable supply chain for both input (battery waste) and output (recovered materials)
- Strong alignment with regulatory and sustainability requirements

Growth Potential

The lithium-ion battery recycling industry in India is at a nascent yet rapidly expanding stage. With increasing EV adoption, government incentives, and regulatory enforcement, the availability of recyclable battery waste is expected to grow significantly. The proposed plant is designed to capitalize on this emerging opportunity with a scalable and future-ready operational model.



4. Project Objectives

The primary objective of the proposed Lithium-Ion Battery Recycling Plant is to establish an efficient, environmentally compliant, and commercially viable system for the recycling and recovery of valuable materials from end-of-life lithium-ion batteries.

Key Objectives

1. Resource Recovery and Value Creation

To recover critical metals such as **Lithium, Cobalt, Nickel, Graphite and Manganese** from battery waste and reintroduce them into the manufacturing value chain, thereby reducing dependence on virgin raw materials.

2. Environmentally Sustainable Waste Management

To provide a scientifically managed solution for handling and processing hazardous lithium-ion battery waste, minimizing environmental impact and promoting sustainable industrial practices.

3. Compliance with Regulatory Framework

To operate in full compliance with **Battery Waste Management Rules, 2022**, CPCB guidelines, and other applicable environmental regulations, enabling producers to meet their Extended Producer Responsibility (EPR) obligations.

4. Development of Integrated Recycling Infrastructure

To establish a facility integrating **mechanical processing (black mass generation)** and **hydrometallurgical processing**, ensuring efficient end-to-end recycling and higher material recovery efficiency.

5. Support to Growing EV and Energy Storage Sector

To contribute to the supply chain of critical materials required for **electric vehicles (EVs), renewable energy storage systems, and electronics manufacturing**, thereby supporting India's clean energy transition.

6. Promotion of Circular Economy

To promote a circular economy model by converting waste into valuable resources and enabling reuse of recovered materials in battery production and allied industries.

7. Safe Handling and Processing of Hazardous Materials

To implement best practices for safe collection, storage, transportation, and processing of lithium-ion batteries, ensuring worker safety and risk mitigation.

8. Creation of Scalable and Future-Ready Operations

To develop a scalable plant design that can accommodate future expansion in capacity and technological upgrades in line with market growth and regulatory developments.

9. Strengthening Industry Linkages

To build strong partnerships with **battery manufacturers, recyclers, aggregators, and EPR stakeholders**, ensuring steady raw material supply and market for recovered products.

10. Contribution to Economic and Industrial Growth

To support the development of a domestic recycling ecosystem, generate employment opportunities, and contribute to the growth of the green and sustainable manufacturing sector.

5. Target Market Segments

Target Market Segments

The proposed Lithium-Ion Battery Recycling Plant will cater to a diverse set of industries and stakeholders across the battery value chain. The target market is driven by both **compliance requirements** and **demand for recovered critical materials**, ensuring strong and sustained business potential.

1. Battery Manufacturers & Giga factories

- Manufacturers of lithium-ion cells and battery packs
- Demand recycled materials such as Lithium, Cobalt, and Nickel for cost optimization and sustainability goals
- Increasing preference for **secondary raw materials** to reduce dependency on imports

2. Electric Vehicle (EV) Manufacturers & OEMs

- EV companies generating production scrap and end-of-life battery waste
- Require recycling partners to meet **EPR obligations**
- Potential for long-term contracts for waste processing and material recovery

3. E-Waste Aggregators & Dismantlers

- Entities involved in collection and primary dismantling of electronic waste
- Supply lithium-ion batteries extracted from devices such as laptops, mobiles, and other electronics
- Key suppliers for consistent raw material inflow

4. Producers & Importers under EPR Framework

- Companies obligated under **Battery Waste Management Rules, 2022**
- Require authorized recyclers to fulfill their recycling targets
- Opportunity for service-based revenue through compliance partnerships

5. Energy Storage & Renewable Sector

- Solar and wind energy storage system providers using lithium-ion batteries
- Increasing volume of battery waste expected over the coming years
- Strategic segment for future raw material sourcing

6. Metal Refining & Chemical Industries

- Buyers of recovered metals and compounds
- Demand high-purity Lithium salts, Cobalt, Nickel intermediates
- Integration into downstream manufacturing and refining processes

7. Industrial Scrap Generators

- Industries generating battery waste during manufacturing or usage
- Includes telecom, data centers, UPS systems, and electronics manufacturing units
- Require safe disposal and recycling solutions

8. Government & Institutional Channels

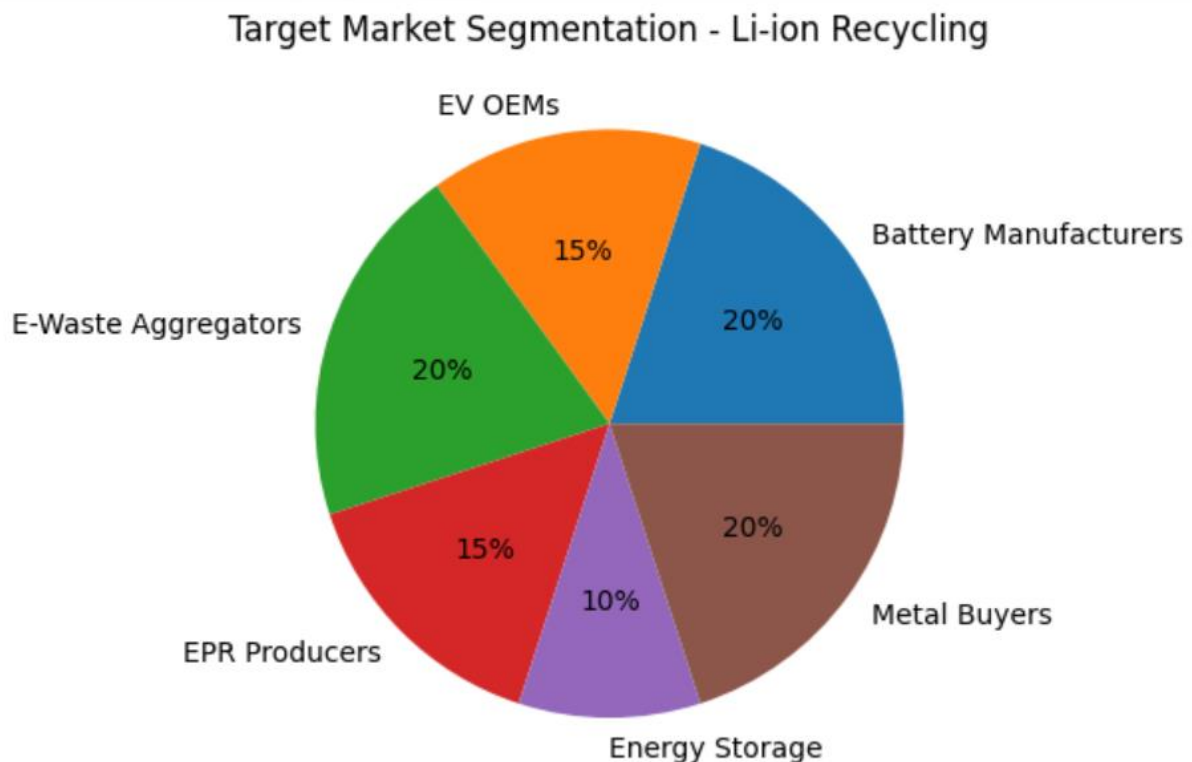
- Public sector undertakings and government-backed initiatives
- Opportunities through tenders, sustainability programs, and recycling mandates

Market Segment Positioning

The plant is strategically positioned to operate across:

- **Supply Side:** Battery waste generators, aggregators, dismantlers
- **Processing Side:** Recycling and material recovery
- **Demand Side:** Manufacturers and industries requiring recovered materials

The target market for the proposed plant is **broad, growing, and regulation-driven**, ensuring both steady supply of raw materials and consistent demand for recovered outputs. By catering to multiple segments across the value chain, the plant can establish a **stable and scalable business model**.



6. Market Analysis

Industry Overview

The lithium-ion battery recycling industry is emerging as a **high-growth and strategically important sector** globally and in India. Driven by the rapid adoption of **electric vehicles (EVs), renewable energy storage systems, and portable electronics**, the volume of end-of-life batteries is increasing significantly. This creates a strong demand for efficient and environmentally compliant recycling infrastructure.

India, being one of the fastest-growing EV markets, is expected to witness a substantial rise in lithium-ion battery consumption, thereby generating a parallel increase in battery waste over the coming years.

Market Drivers

1. Growth of Electric Vehicles (EVs)

- Rapid EV adoption under government initiatives such as **FAME-II**
- Increasing battery usage in two-wheelers, three-wheelers, and passenger vehicles
- Expected surge in battery replacement demand in the next 3–5 years

2. Regulatory Push (Battery Waste Management Rules, 2022)

- Mandatory recycling targets for producers and importers
- Strict enforcement of **Extended Producer Responsibility (EPR)**
- Growing demand for authorized recyclers to fulfil compliance obligations

3. Rising Demand for Critical Metals

- Lithium, Cobalt, and Nickel are essential for battery manufacturing
- Limited domestic availability → heavy reliance on imports
- Recycling provides a cost-effective and sustainable alternative

4. Increasing Electronic Waste

- High consumption of smartphones, laptops, and consumer electronics
- Continuous generation of lithium-ion battery waste from e-waste stream

5. Sustainability & Circular Economy Focus

- Strong push from industries and governments towards **green manufacturing**
- Preference for recycled materials by global OEMs

Market Size & Growth Potential

- The lithium-ion battery recycling market in India is currently in a **nascent stage but expanding rapidly**
- Expected to grow at a **high CAGR over the next 5–10 years**
- Increasing investments from both startups and large industrial players
- Significant untapped potential due to limited formal recycling capacity

Demand-Supply Gap

- Current recycling capacity in India is **limited compared to future waste generation**
- Large volume of batteries is either:
 - Unprocessed
 - Handled by informal sector (environmental risk)
- Strong opportunity for organized players with **compliant and scalable operations**

Competitive Landscape

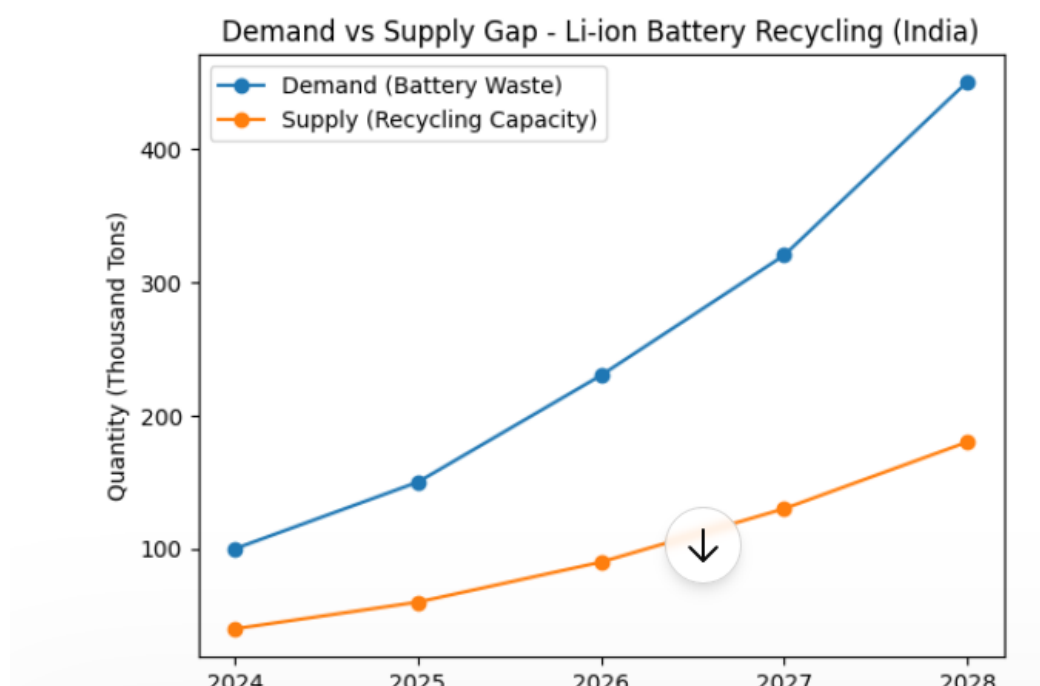
The market is witnessing entry of:

- **Organized recyclers and startups** focusing on advanced technologies
- **Large industrial groups** entering battery recycling space
- **Global collaborations and technology partnerships**

However, the industry still lacks:

- Sufficient processing capacity
- Standardized recovery efficiency
- Strong nationwide collection networks

This creates a favourable environment for new entrants with integrated capabilities.



Challenges in the Market

- Fragmented collection and supply chain
- Dependence on informal sector for raw material
- Technology selection and recovery efficiency challenges
- Regulatory approvals and compliance requirements
- Price volatility of recovered metals

Opportunities

1. EPR-Driven Business Model

- Long-term contracts with producers and importers
- Stable demand due to compliance requirements

2. Integration across Value Chain

- Opportunity to integrate collection, recycling, and material supply
- Higher margins through downstream processing

3. Export Potential

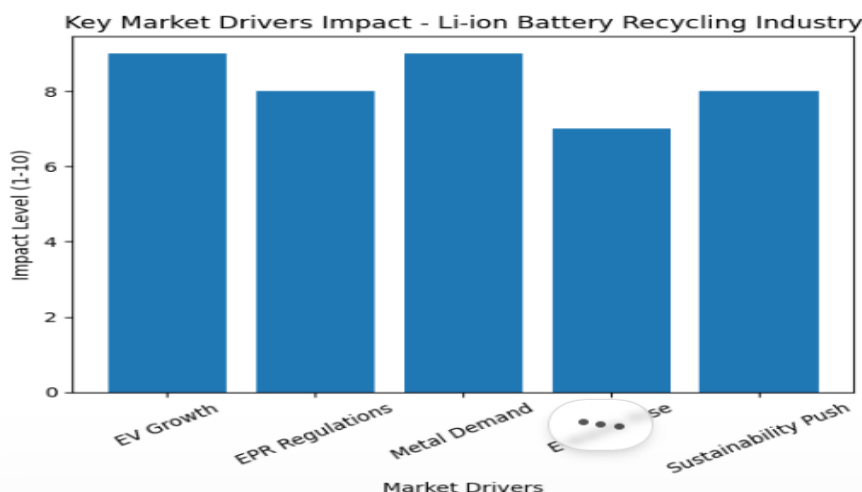
- Demand for black mass and refined metals in international markets

4. Strategic Partnerships

- Collaboration with EV manufacturers, OEMs, and battery companies
- Technology tie-ups for improved recovery rates

Future Outlook

The lithium-ion battery recycling industry is expected to become a **core component of India's energy and mobility ecosystem**. With increasing regulatory enforcement, rising battery consumption, and global sustainability commitments, the sector is projected to witness **exponential growth**. The market presents a **strong, regulation-driven, and future-ready opportunity** with significant growth potential. The combination of rising battery waste, increasing demand for recovered materials, and limited organized recycling infrastructure positions the proposed plant favourably to capture market share and establish a sustainable business presence.



7. Subsidies and Incentives

5 TPD Lithium-Ion Battery Recycling + 0.5 TPD Hydrometallurgy Plant

Proposed capacity

Section	Daily capacity	Annual capacity at 300 days
Battery mechanical recycling	5 TPD	1,500 TPA battery scrap
Black-mass output, assuming 40% yield	2 TPD	600 TPA
Hydrometallurgical processing	0.5 TPD	150 TPA black mass
Surplus black mass available for sale	Around 1.5 TPD	Around 450 TPA

The project should be described as an **integrated lithium-ion battery recycling and critical-mineral recovery unit**, rather than only a black-mass facility. The hydrometallurgy section supports the value-addition and circular-economy positioning, although its proposed scale is relatively small for the principal Central critical-mineral incentive.

1. Central Government Support Applicable in Both States

A. CGTMSE collateral-free credit guarantee

CGTMSE guarantee coverage was increased from ₹5 crore to ₹10 crore for eligible credit facilities approved on or after 1 April 2025.

CGTMSE does not directly lend money or provide a capital subsidy. It gives a guarantee to the member lending institution, enabling the bank to sanction eligible term loan and working-capital facilities without full collateral security. Final sanction remains subject to the bank's appraisal of promoter contribution, cash flows, and credit history and project viability.

Practical structure for this project

Particular	Illustrative amount
Total project cost	₹8.00 crore
Promoter contribution, assumed 25%	₹2.00 crore
Term loan and working capital	₹6.00 crore
CGTMSE-covered portion	Up to ₹6.00 crore, subject to bank approval

Particular	Illustrative amount
Collateral requirement	May be waived for the covered portion
Personal guarantee	May still be required by the lender
Guarantee fee	Payable as per applicable CGTMSE slab

A project requiring only ₹5 crore of bank finance can still seek coverage for the full eligible ₹5 crore. Under the hybrid-security option, a bank may take collateral for one part of the facility and obtain CGTMSE cover for the unsecured part.

Key CGTMSE documents

- Udyam registration showing Micro or Small Enterprise status
- Company incorporation, PAN and GST
- Detailed project report
- Five- to seven-year financial projections
- Machinery quotations
- Land ownership or registered lease
- Promoter net-worth statement
- Income-tax returns and bank statements
- CIBIL and credit-history documents
- CTE or proof of environmental application
- Means-of-finance statement
- Projected DSCR, break-even and cash-flow analysis
- Raw-material supply and product-offtake arrangements

Important: CGTMSE is a credit guarantee, not a grant. The borrower remains fully liable to repay the loan.

B. MSME ZED and quality-certification support

The unit may apply for MSME quality and sustainability programmes such as:

- ZED certification
- ISO 9001
- ISO 14001
- ISO 45001
- Testing and product-quality certifications

These are smaller reimbursements and capacity-building benefits, not major project-capital subsidies.

2. Haryana — Subsidies and Incentives

A. Policy position

The **Make in Haryana Industrial Policy 2025** differentiates incentives by:

- Enterprise size
- Development-block category
- Investment level
- Sector classification
- Employment and sustainability commitments

The policy provides that total state incentives cannot exceed 100% of fixed capital investment and that eligible state incentives may be over and above Central Government incentives.

For a plant of this scale, the unit is likely to fall in the MSME segment, subject to its final investment and turnover. The exact package cannot be fixed until the proposed district, development block, project cost and final sector classification are known.

B. Incentive heads to be pursued in Haryana

Incentive head	Potential applicability
Capital investment assistance	Subject to notified MSME scheme, block and sector
Net SGST-linked assistance	Subject to eligibility and operational scheme
Interest subsidy	Possible on eligible term loan
Electricity-duty assistance	Location and scheme dependent
Stamp-duty assistance	Depends on block and land transaction
Employment-generation support	Possible for qualifying local employment
Green technology assistance	Relevant for ETP, APCS, pollution control and resource efficiency
Quality certification assistance	Possible for ISO/ZED and prescribed certifications
R&D or technology support	Case-specific for advanced recycling technology

C. Haryana strategy

You can obtain an **in-principle incentive eligibility confirmation** through Invest Haryana before:

1. Finalizing the land;
2. Placing machinery purchase orders;
3. Starting construction;
4. Commencing commercial production.

This is important because the level of benefit changes materially between more industrialized NCR blocks and comparatively less-developed blocks.

D. Indicative Haryana financial treatment

Until the exact block and operational scheme are confirmed, the DPR should use:

Item	Base-case treatment
Capital subsidy	Nil in base financial model
Interest subsidy	Nil in base case; show separately as potential upside
SGST reimbursement	Nil in base case
Stamp/electricity benefits	Disclose but do not capitalize
CGTMSE	Include as financing route, not revenue or subsidy

This prevents overstating project returns before receipt of a written eligibility confirmation.

E. Haryana advantages

- Better access to Delhi-NCR battery scrap, OEMs and aggregators
- Easier technical-manpower availability
- Strong logistics and downstream buyer network
- Better availability of laboratories and environmental consultants
- Lower inbound logistics risk

F. Haryana limitations

- Land and rental costs are generally higher near NCR
- Less-developed-block incentives may require locating farther from the principal raw-material markets
- Environmental scrutiny for battery storage and hazardous operations can be stringent

3. Uttarakhand — Subsidies and Incentives

The Uttarakhand MSME Policy 2023 offers a more clearly defined capital-subsidy structure. The benefit depends on the district/area category and eligible fixed capital investment.

A. Location categories

Category	Principal areas
A	Pithoragarh, Uttarkashi, Chamoli, Champawat, Rudraprayag and Bageshwar
B	Almora, Pauri Garhwal and specified hill areas
C	Specified semi-hill areas of Dehradun, Tehri and Nainital
D	Haridwar, Udham Singh Nagar and principal plain industrial areas of Dehradun and Nainital

The policy treats general eligible MSME manufacturing activities as permissible unless they fall in the prohibited list.

B. Capital subsidy

For eligible plant-and-machinery investment above ₹1 crore and up to ₹5 crore

Uttarakhand category	Capital subsidy formula	Maximum
A	₹50 lakh + 25% of investment above ₹1 crore	₹1.50 crore
B	₹40 lakh + 20% of investment above ₹1 crore	₹1.20 crore
C	₹30 lakh + 12.5% of investment above ₹1 crore	₹80 lakh
D	₹20 lakh + 10% of investment above ₹1 crore	₹60 lakh

Example: ₹5 crore eligible fixed capital investment

Location category Indicative capital subsidy

Category A	₹1.50 crore
Category B	₹1.20 crore
Category C	₹80 lakh
Category D	₹60 lakh

This calculation applies to **eligible fixed capital investment**, not necessarily the entire project cost. Land, land development, working capital, pre-operative expenses, consultancy fees and several administrative assets are excluded. The policy does, however, include directly related machinery, eligible workshop buildings, pollution-control equipment, transformers, boilers, DG sets, installation, transportation and certain manufacturing-support systems.

C. Additional capital subsidy

Where the lithium-ion battery recycling activity is formally accepted under a priority or most-priority category, an additional benefit may apply:

Qualification	Additional rate	Small-enterprise ceiling
Most-priority activity in Category A or B	10%	₹15 lakh
Most-priority activity in Category C or D	5%	₹10 lakh
Priority category activity	5%	₹10 lakh
Eligible SC/ST/women/disabled-owned unit	5%	₹10 lakh

The activity's priority classification must be confirmed in writing before including this benefit in the financial model.

D. Interest subsidy

For a new **small manufacturing enterprise**, the policy provides:

Location category	Interest subsidy	Annual ceiling	Maximum period
A	3%	₹6 lakh/year	3 years
B	3%	₹5 lakh/year	3 years
C	3%	₹4 lakh/year	3 years
D	3%	₹3 lakh/year	3 years

The assistance applies to eligible term loans for workshop buildings and plant and machinery. It does not apply to land, land development, working capital or cash-credit limits.

Maximum three-year benefit for a small enterprise

Category Maximum total interest reimbursement

A	₹18 lakh
B	₹15 lakh
C	₹12 lakh
D	₹9 lakh

E. Stamp-duty reimbursement

Category Reimbursement

A	100%
B	100%
C	75%
D	50%

This is subject to the policy's conditions concerning acquisition or lease of land and buildings.

F. Electricity-duty exemption

New eligible MSMEs with sanctioned electrical load up to **500 kW** may receive electricity-duty reimbursement for **five years**. A 5 TPD mechanical recycling line with a small hydromet section may be kept within this limit depending on the final technology and utility design.

G. Quality certification support

Uttarakhand provides reimbursement of:

75% of eligible certification expenditure, subject to a maximum of ₹1 lakh per unit.

It may cover eligible costs relating to ISO, BIS, pollution-control certification, patents, trademarks and ZED certification.

I. Important Uttarakhand conditions

You must:

- Apply through the Single Window CAF before commercial production;
- Obtain in-principle approval before commencement;
- Use new machinery;
- Obtain CTO and all required statutory approvals;
- Make machinery payments through banking channels against GST invoices;
- Maintain at least 70% permanent employment for Uttarakhand residents;
- Submit the capital-subsidy claim within the prescribed period after commercial production.

4. Indicative Uttarakhand Subsidy Package

Assuming:

- Eligible fixed capital investment: ₹5 crore
- Unit classification: Small enterprise
- New unit
- Electrical load below 500 kW
- All policy conditions satisfied

Benefit	Category A	Category B	Category C	Category D
Capital subsidy	₹1.50 Cr	₹1.20 Cr	₹0.80 Cr	₹0.60 Cr
Maximum interest subsidy over three years	₹0.18 Cr	₹0.15 Cr	₹0.12 Cr	₹0.09 Cr
Stamp-duty reimbursement	100%	100%	75%	50%
Electricity-duty benefit	5 years	5 years	5 years	5 years
Certification reimbursement	Up to ₹1 lakh	Up to ₹1 lakh	Up to ₹1 lakh	Up to ₹1 lakh
Quantifiable benefit excluding stamp and electricity duty	₹1.69 Cr	₹1.36 Cr	₹0.93 Cr	₹0.70 Cr

Additional priority-category assistance may increase the package, but it should be included only after formal activity classification.

5. Haryana vs Uttarakhand

Factor	Haryana	Uttarakhand
Subsidy calculation certainty	Depends strongly on notified scheme and block	Clearly structured under MSME Policy 2023
Capital-subsidy potential	To be confirmed	Up to ₹1.50 crore on ₹5 crore eligible FCI in Category A
Raw-material availability	Stronger	Weaker; likely dependent on NCR and other states
Customer and OEM access	Stronger	Moderate
Land cost	Higher near NCR	Lower in several industrial areas
Skilled manpower	Stronger near NCR	Moderate

Factor	Haryana	Uttarakhand
Logistics cost	Lower for NCR scrap	Higher if sourcing from NCR
Best practical locations	Non-NCR incentivized blocks with road access	Haridwar or Udham Singh Nagar for operations; hill categories for higher subsidy but harder logistics

6. Recommendation

For maximum subsidy

Uttarakhand Category A or B provides the strongest quantified capital support. However, locating a hazardous battery-processing operation in a remote hill district may create major feedstock, logistics, skilled-manpower and emergency-response constraints.

For a practical subsidy-plus-logistics balance

Haridwar or Udham Singh Nagar, Uttarakhand, is the more practical Uttarakhand option. These are Category D areas, so the capital subsidy is lower, but industrial infrastructure and access to NCR are better.

For raw-material security and easier market access

Haryana is stronger, particularly where the client can secure a suitable industrial location close to NCR. Before selection, a written incentive assessment should be obtained for the exact development block.

Recommended funding structure

- Promoter contribution: 25%–30%
- CGTMSE-backed term loan and working capital: 70%–75%
- State subsidy treated as post-commissioning reimbursement
- Critical Mineral Recycling Scheme excluded from base projections
- EPR certificate revenue excluded from debt-servicing calculations unless supported by executed agreements

The safest commercial approach is to compare an identified Haryana site against **Haridwar/Udham Singh Nagar** using actual land cost, transport cost, power tariff, eligible machinery value and written state incentive confirmation.

8. Land feasibility of Haryana and Uttarakhand

Location Assessment

5 TPD Lithium-Ion Battery Recycling and 0.5 TPD Hydrometallurgy Plant

Considering that you are based in Punjab, the three practical location options are:

1. **Kundli–Rai industrial belt, Sonipat**
2. **Panipat–Samalkha/Barhi industrial belt**
3. **Roorkee–Bhagwanpur–Haridwar industrial corridor**

The project should preferably be established only in a **notified or government-approved industrial estate** where the land-use permission, industrial infrastructure, power connection, fire access and pollution-control approvals are easier to demonstrate.

Overall Recommendation

Best balanced location: Bhagwanpur–Roorkee industrial area, Uttarakhand

The **Bhagwanpur–Roorkee–Haridwar corridor** offers the best overall combination of:

- Access from Punjab and western Uttar Pradesh
- Proximity to Delhi-NCR battery-waste markets
- Existing industrial infrastructure
- More clearly defined MSME capital incentives
- Lower land and rental pressure than Kundli
- Availability of industrial plots and sheds around established industrial clusters
- State single-window and district-level investor facilitation

The Roorkee/Bhagwanpur area falls in Haridwar district, which is treated as a **Category D area** under the Uttarakhand MSME framework. Although this offers a lower subsidy than remote hill categories, it provides a much better operational balance in terms of logistics, manpower and industrial infrastructure. Uttarakhand's MSME policy provides clearly structured capital, interest, stamp-duty and electricity-duty benefits for eligible manufacturing enterprises.

1. Kundli–Rai, Sonipat

Location profile

Suitable zones include:

- Kundli Industrial Area
- HSIIDC Rai Industrial Estate
- Areas adjoining the Kundli–Rai–Narela industrial corridor
- Approved industrial estates near KMP and NH-44

Haryana operates an integrated investor-facilitation system through Invest Haryana and HSIIDC, which assists industries with land, building-plan approvals and statutory clearances. (HSIIDC)

Advantages

Raw-material availability: Excellent

Kundli/Rai has the strongest access to:

- Delhi-NCR e-waste aggregators
- Battery pack assemblers
- EV service networks
- Consumer-electronics battery scrap
- Industrial scrap markets
- Noida, Ghaziabad, Delhi and Gurugram suppliers

This is the strongest option where the priority is uninterrupted raw-material supply.

Market access: Excellent

Finished black mass and recovered metals can be transported easily to:

- Delhi-NCR refiners and traders
- Haryana industrial consumers
- Punjab
- Western Uttar Pradesh
- Rajasthan

Logistics

Kundli is located on the Delhi–Punjab transport corridor and provides convenient access through NH-44 and the KMP network.

Limitations

Incentive availability

Kundli and Rai are highly industrialized areas. Haryana differentiates incentives according to development-block category, with stronger fiscal benefits generally directed toward less-developed locations. The Make in Haryana Industrial Policy 2025 continues the block-category approach and offers differentiated benefits based on location and sector. ([Invest Haryana](#))

The project should not assume a major capital subsidy at Kundli without written confirmation from Invest Haryana.

Environmental scrutiny

Kundli/Sonipat is within the broader NCR regulatory environment. A lithium-ion battery recycling unit involving shredding, black-mass dust, chemicals and hydrometallurgy would face:

- Strict air-emission scrutiny
- CAQM-related operational conditions
- Higher expectations for enclosed processing
- Full dust extraction and bag filtration
- Scrubber systems
- ETP/ZLD for hydrometallurgy
- Fire and thermal-runaway controls

Land cost

Industrial land and compliant sheds are likely to be relatively expensive because of proximity to Delhi.

Suitability score

Factor	Score
Raw-material availability	5/5
Market access	5/5
Punjab connectivity	5/5
Ease of environmental approval	2.5/5
Incentive potential	2.5/5
Land affordability	2/5
Overall	3.7/5

Verdict

Kundli–Rai is best where **raw-material security and NCR access outweigh subsidy considerations**.

2. Panipat–Samalkha–Barhi Belt

Preferred zones

- HSIIDC industrial areas around Panipat
- Samalkha industrial belt
- Barhi industrial estate
- Approved industrial areas with proper industrial land use and road access

Panipat has a significant manufacturing and MSME base, and Haryana’s industrial policy recognizes Panipat as an established industrial centre.

Advantages

Punjab connectivity: Excellent

Panipat is directly accessible from Punjab through NH-44 and is more convenient for the promoter than most NCR locations.

Raw-material availability: Good

Potential procurement catchment includes:

- Punjab
- Chandigarh–Mohali–Ludhiana corridor
- Haryana
- Delhi–NCR
- Western Uttar Pradesh

It is less dependent on Delhi alone than Kundli.

Industrial ecosystem

Panipat has:

- Industrial workforce
- Transport operators
- Fabrication and maintenance vendors
- Warehousing
- Power and industrial infrastructure
- Existing chemical and manufacturing support services

Land cost

Land or shed rental is likely to be more manageable than Kundli, particularly outside the premium industrial pockets.

Limitations

NCR-related environmental controls

Panipat is also part of the NCR planning and air-quality regulatory ecosystem. A battery recycling and hydrometallurgy facility will therefore still require robust environmental controls.

Incentives

Panipat is an established industrial district. The exact Haryana block classification of the selected village or industrial estate must be checked before concluding the subsidy package. Haryana incentives vary by block and may include SGST-linked assistance, interest subsidy, electricity-duty exemption and stamp-duty support, but entitlement requires formal confirmation.

Suitability score

Factor	Score
Raw-material availability	4/5
Market access	4/5
Punjab connectivity	5/5
Ease of environmental approval	3/5
Incentive potential	3/5
Land affordability	3.5/5
Overall	3.8/5

Verdict

Panipat is the **best Haryana option** because it offers better balance than Kundli between raw-material access, Punjab connectivity, land cost and industrial support.

3. Bhagwanpur–Roorkee–Haridwar Corridor

Preferred micro-locations

You should examine:

1. **Bhagwanpur Industrial Area**
2. Industrial estates between Bhagwanpur and Roorkee
3. **IIE Haridwar / SIDCUL industrial areas**
4. Existing compliant industrial sheds in Haridwar district
5. Private approved industrial estates listed through Invest Uttarakhand

SIIDCUL develops and manages industrial infrastructure in Uttarakhand, including industrial estates in the Haridwar region. The state also operates a single-window investor platform and district-level Udyog Mitra mechanism for resolving investor issues.

Advantages

Incentive certainty: Best of the three

For an eligible small manufacturing enterprise with eligible fixed capital investment between ₹1 crore and ₹5 crore, Category D areas can receive:

- Capital subsidy under the applicable formula, subject to a maximum of ₹60 lakh
- Interest subsidy of 3%, subject to ₹3 lakh per year for three years
- Stamp-duty reimbursement up to 50%
- Electricity-duty reimbursement for eligible units with load up to 500 kW for five years
- Quality-certification reimbursement, subject to scheme limits

The final eligible amount depends on enterprise classification, admissible fixed capital investment and timely application.

Ease of approvals: Good

Approvals are generally more manageable when the unit is located in a recognized industrial estate with:

- Industrial land use
- Road and drainage infrastructure
- Power connectivity
- Fire access
- Approved building use
- Dedicated industrial administration

Raw-material catchment: Good

The plant can procure from:

- Punjab
- Chandigarh and Mohali
- Dehradun and Haridwar
- Saharanpur and western Uttar Pradesh
- Delhi-NCR
- Industrial battery and electronics users in Uttarakhand

It is not as close to Delhi as Kundli, but it has a broader multi-state procurement catchment.

Proximity to promoter

Roorkee is reasonably accessible from Punjab through the Ambala–Yamunanagar–Saharanpur route, making management and supervision practical.

Lower location pressure

Industrial rental and land costs are generally more manageable than the Kundli–Delhi border belt.

Limitations

Lower local battery-waste concentration

Haridwar/Roorkee does not have the same concentrated battery and e-waste aggregation network as Delhi-NCR. The client will need:

- Delhi-NCR procurement agreements
- Punjab collection partners
- OEM and battery-assembler tie-ups
- Authorized transport arrangements
- 15–20 days of feedstock inventory

Transport cost

Material sourced from Delhi-NCR or Punjab will incur higher inbound freight than a Kundli plant.

Subsidy classification

Roorkee/Haridwar is Category D, so it does not receive the much larger Category A or B hill-area subsidy. However, Category D is operationally more suitable for a battery recycling facility than remote hill districts.

Suitability score

Factor	Score
Raw-material availability	3.5/5
Market access	4/5
Punjab connectivity	4/5
Ease of environmental approval	4/5
Incentive potential	4.5/5
Land affordability	4/5
Overall	4.0/5

Verdict

Bhagwanpur–Roorkee is the **best overall location**, provided the client secures long-term battery-scrap supply agreements before commissioning.

Comparative Ranking

Rank	Location	Best feature	Main concern
1	Bhagwanpur–Roorkee	Incentives, industrial infrastructure and approval environment	Feedstock must be contracted from multiple states
2	Panipat–Samalkha/Barhi	Best Haryana balance and Punjab access	Lower subsidy certainty and NCR compliance
3	Kundli–Rai	Best raw-material and customer access	High land cost and strict NCR environmental scrutiny

Recommended Site Strategy

Primary recommendation

Select an existing industrial shed or industrial plot in:

Bhagwanpur Industrial Area, near Roorkee, Haridwar district

The site should preferably be:

- Within an approved industrial estate
- At least 3,000–4,000 sq. m for the stated capacity
- Accessible to fire tenders and heavy vehicles

- Equipped with adequate sanctioned power
- Suitable for chemical storage and an ETP
- Away from residential habitation, schools and hospitals
- Outside flood-prone and environmentally restricted zones
- Permitted for the proposed industry category

Haryana fallback

Where raw-material supply is the dominant consideration, select:

An approved industrial estate near Panipat or Samalkha rather than Kundli

This provides better access to Punjab while retaining reasonable connectivity to Delhi-NCR.

Raw-Material Procurement Plan by Location

Bhagwanpur–Roorkee

Source region	Target share
Punjab and Chandigarh	25%
Delhi-NCR	30%
Western Uttar Pradesh	25%
Uttarakhand	10%
OEM/manufacturing scrap contracts	10%

Panipat

Source region	Target share
Punjab	30%
Delhi-NCR	30%
Haryana	20%
Western Uttar Pradesh	10%
OEM/manufacturing contracts	10%

Pre-Land-Finalization Checklist

Before executing a sale deed or long lease, obtain written confirmation on:

1. Industrial land-use permissibility.
2. Whether lithium-ion battery recycling and hydrometallurgy are permitted at the site.
3. MP/HP power availability and transformer requirement.
4. Groundwater and water-supply availability.
5. Fire-access road and setback compliance.
6. Distance from habitation and sensitive receptors.
7. Flood-zone and drainage status.
8. Availability of authorized hazardous-waste disposal facility.
9. Eligibility under the relevant state incentive policy.
10. Whether subsidy application must be filed before machinery ordering or commercial production.

Final Recommendation

For you, **Bhagwanpur–Roorkee should be the first-choice location** because it provides the best combined position on incentives, approval facilitation, industrial infrastructure, promoter access from Punjab and manageable operating costs.

Panipat–Samalkha should be retained as the second option where you value raw-material availability and Punjab proximity more than capital subsidy.

Kundli should be selected only when you have strong Delhi-NCR supply contracts and can absorb higher land cost and stricter NCR environmental compliance.

9. Product and service offerings

1. Product Offerings (Primary Revenue Streams)

These come from both **mechanical processing (black mass)** and **hydrometallurgical refining**.

A. Intermediate Product

- **Black Mass (5 TPD capacity)**
 - Contains: Lithium, Cobalt, Nickel, Manganese, Graphite
 - Sold to:
 - Refiners
 - Cathode manufacturers

B. Refined Battery Materials (Hydrometallurgy – 0.5 TPD)

Higher-margin products:

1. Lithium Products

- Lithium Carbonate (Li_2CO_3)
- Lithium Hydroxide (LiOH) (if upgraded later)

2. Cobalt Products

- Cobalt Sulphate (CoSO_4)
- Cobalt Oxide (Co_3O_4)

3. Nickel Products

- Nickel Sulphate (NiSO_4)

4. Manganese Products

- Manganese Sulphate (MnSO_4)

C. By-products / Secondary Materials

- Copper scrap (from current collectors)
- Aluminum scrap
- Steel fractions
- Graphite powder

- Plastic separators (low-value recovery)

D. Future Product Expansion

- Precursor Cathode Active Material (pCAM)
- Direct cathode recycling
- Battery-grade chemicals

2. Service Offerings (Recurring Revenue Model)

These are critical for steady cash flow and long-term contracts.

A. Battery Recycling Services

- Collection and recycling of:
 - EV batteries
 - E-waste batteries (laptops, phones)
 - Industrial batteries
- Charging:
 - Per ton processing fee
 - Or revenue-sharing model

B. Extended Producer Responsibility (EPR) Services

- Help OEMs comply with **Battery Waste Management Rules (India)**
- Services include:
 - Collection network setup
 - Recycling certification
 - Documentation & compliance reporting

C. Toll Processing / Contract Recycling

- Process batteries or black mass for:
 - OEMs
 - Importers
 - Traders
- Client retains ownership of recovered metals

D. Secure Disposal Services

- Safe handling of:
 - Defective batteries
 - Recall batches
- Important for EV companies

E. Logistics & Collection Services

- Battery pickup and transportation
- Reverse logistics network

F. Data Destruction & Certification

- Certification of:
 - Battery destruction
 - Material recovery
- Required by corporates & OEMs

3. Integrated Business Model

Revenue Mix

- 80–90% → Sale of refined metals
- 5–7% → Recycling services & EPR contracts
- 2–3% → Scrap/by-products

4. Strategic Positioning

Highlight:

- **Dual revenue model** (products + services)
- Ability to shift between:
 - Selling black mass OR
 - Refining in-house for higher margins
- Alignment with:
 - EV growth
 - Government recycling mandates

10. Raw material and Procurement plan

1. Raw Material Requirement

◆ A. Types of Raw Materials

Primary Input

- End-of-life Lithium-ion batteries:
 - EV batteries (2W, 3W, 4W)
 - Consumer electronics batteries (laptop, mobile)
 - Industrial/storage batteries

Secondary Input

- Battery scrap:
 - Manufacturing rejects (gigafactories)
 - Cathode/anode scrap
- Imported black mass (optional for hydromet plant)

B. Quantity Requirement

Section	Input Requirement
Mechanical plant	~5 TPD batteries
Hydromet plant	~0.5 TPD black mass
Conversion ratio	~30–50% black mass yield

To support 0.5 TPD hydromet:

- Need 2-2.5 TPD battery input

2. Procurement Sources

A. OEM & EV Companies (High Priority)

- Electric vehicle manufacturers
- Battery manufacturers
- Energy storage companies

Strategy:

- Long-term contracts
- EPR compliance partnerships

B. E-Waste Aggregators / Dealers

- Informal and formal sector collectors
- Scrap traders

Strategy:

- Regional aggregation network
- Fixed-rate procurement agreements

C. Authorized Collection Centers

- Set up own:
 - Collection points in cities
- Partner with:
 - E-waste companies

D. Import of Black Mass (Optional)

- From:
 - Europe
 - US
 - Southeast Asia

Useful when:

- Domestic supply is low
- Hydromet plant needs steady feed

E. Battery Manufacturers (Scrap Supply)

- Production scrap
- Defective cells

3. Procurement Model

◆ A. Purchase Model

- Buy batteries at ₹/kg
- Typical (indicative):

- ₹80–₹370/kg (depends on chemistry)

◆ B. Tolling Model

- Client provides batteries
- You charge:
 - ₹/ton processing fee

C. EPR Model (Most Important)

- OEM pays for recycling
- You earn:
 - Processing fee + material value

4. Logistics & Supply Chain

A. Transportation

- Hazardous material handling vehicles
- Fire-safe packaging

B. Storage Requirements

- Fireproof battery storage area
- Segregation:
 - Damaged batteries
 - Different chemistries

C. Inventory Planning

Parameter	Value
Buffer stock	7–15 days
Storage capacity	50–100 tons

5. Procurement Strategy

◆ Phase 1 (0–1 Year)

- Tie-ups with:
 - Local scrap dealers
 - Small OEMs
- Focus: volume buildup

◆ Phase 2 (1–3 Years)

- Long-term contracts with:
 - EV OEMs
 - Battery companies
- Develop collection network

◆ Phase 3 (3+ Years)

- Direct sourcing from:
 - Gigafactories
- Import black mass if needed

6. Risk Mitigation

Risk	Mitigation
Supply shortage	Multi-source procurement
Price fluctuation	Long-term contracts
Quality variation	Sorting & testing system
Regulatory risk	EPR compliance partnerships

7. Quality Control at Procurement

- Battery chemistry identification (LFP, NMC, etc.)
- Moisture control
- Removal of non-recyclables
- Testing for metal content

8. Cost Consideration

- Raw material = **60–70% of total operating cost**
- Key driver of profitability

The raw material for the proposed plant will primarily consist of end-of-life lithium-ion batteries sourced from EV manufacturers, battery producers, and e-waste aggregators. The company will establish long-term supply agreements under the Extended Producer Responsibility (EPR) framework to ensure a steady supply of feedstock. A diversified procurement strategy, including domestic sourcing and potential import of black mass, will be adopted to mitigate supply risks. Adequate storage and logistics infrastructure will be developed to handle hazardous materials safely and efficiently.

11. Technology and Equipment used

1. Technology Overview

The proposed plant adopts a **two-stage recycling technology**:

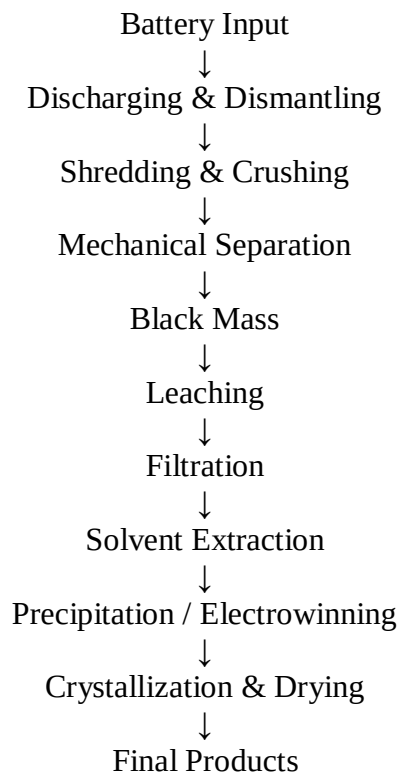
Stage 1: Mechanical Processing (5 TPD)

- Objective: Safe dismantling and separation of battery components
- Output: **Black Mass + Cu, Al, Fe fractions**

Stage 2: Hydrometallurgical Processing (0.5TPD)

- Objective: Recovery of high-value metals
- Output:
 - Lithium compounds
 - Cobalt salts
 - Nickel salts
 - Manganese salts

2. Process Flow Overview



3. Mechanical Processing Technology (Black Mass Plant)

A. Technology Description

- Fully automated **physical separation process**
- Operates under controlled conditions (dust & safety control)
- Multi-stage separation for high purity

B. Key Equipment

1. Feeding & Handling

Equipment	Function
Feeding conveyor	Controlled battery feeding
Hopper & storage bins	Raw material handling

2. Size Reduction

Equipment	Function
Double shaft shredder	Primary size reduction
Hammer crusher	Secondary crushing
Grinding mill	Fine powder generation

3. Separation System

Equipment	Function
Zigzag separator	Shell/plastic removal
Magnetic separator	Iron removal
Vibrating screens	Size classification
Airflow separator	Cu–Al separation

4. Material Handling

Equipment	Function
Cyclone separators	Powder collection
Bucket elevator	Vertical transport
Conveyors	Material transfer

5. Pollution Control (ECS)

Equipment	Function
Dust collectors (bag filters)	Dust removal
Cyclone system	Pre-cleaning
Spray tower	Gas treatment
Activated carbon unit	VOC removal

C. Output

- Black mass (~60%)
- Copper
- Aluminum
- Steel scrap

4. Hydrometallurgical Technology

A. Technology Description

- **Acid leaching + solvent extraction + precipitation**
- High metal recovery efficiency
- Closed-loop chemical system

B. Process Stages & Equipment

1. Leaching Section

Equipment	Function
SS reactors with agitators	Acid leaching
Chemical dosing system	Controlled reaction
Slurry tanks	Mixing & holding

2. Solid-Liquid Separation

Equipment	Function
Filter press	Separation of residue
Settling tanks	Clarification

3. Purification & Precipitation

Equipment	Function
Neutralization tanks	Impurity removal
Precipitation reactors	Metal recovery

4. Solvent Extraction (SX)

Equipment	Function
Mixer-settlers	Metal separation
Organic tanks	Solvent storage
Pumps & piping	Fluid transfer

5. Electrowinning (Selective)

Equipment	Function
Electrolytic cells	Metal recovery
Rectifiers	Power supply
Cathodes/anodes	Deposition

6. Evaporation & Crystallization

Equipment	Function
------------------	-----------------

Evaporators	Concentration
-------------	---------------

Crystallizers	Salt formation
---------------	----------------

Centrifuge	Solid separation
------------	------------------

Dryers	Final drying
--------	--------------

5. Utility Equipment

A. Power System

Equipment	Function
------------------	-----------------

Transformer	Power supply
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DG set	Backup power
--------	--------------

MCC/PLC panels	Control system
----------------	----------------

B. Water System

Equipment	Function
------------------	-----------------

RO/DM plant	Process water
-------------	---------------

Storage tanks	Water supply
---------------	--------------

C. Heating System

Equipment	Function
------------------	-----------------

Boiler / thermic fluid heater	Steam generation
-------------------------------	------------------

D. Cooling System

Equipment	Function
------------------	-----------------

Cooling tower	Heat removal
---------------	--------------

Chiller (optional)	Temperature control
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E. Compressed Air

Equipment	Function
Air compressor	Pneumatic systems

6. Environmental Control Systems

◆ **A. Effluent Treatment (ETP)**

Equipment	Function
Neutralization tank	pH correction
Clarifier	Solid separation
Filter press	Sludge removal
Evaporator (ZLD)	Zero discharge

◆ **B. Air Pollution Control**

Equipment	Function
Cyclone separator	Dust removal
Bag filter	Fine particle control
Scrubber	Gas cleaning
Carbon filter	Odor removal

7. Automation & Control

- PLC-based process control
- SCADA monitoring
- Sensors:
 - pH
 - Temperature
 - Flow
 - Pressure

12. List of machineries

Black Mass- Metal separation

1. Plant Overview

- **Capacity:** 400–500 kg/hour (~10–12 TPD if worked in 2 shifts)
- **Total Power:** ~299.7 kW
- **Process Type:** Fully **physical/mechanical separation**
- **Output Products:**
 - Black mass
 - Copper granules
 - Aluminum granules
 - Plastic
 - Iron

2. Complete Machinery List

◆ A. Feeding & Primary Handling

Sr.	Machine	Function	Power
1	Feeding Belt Conveyor (covered)	Feed batteries to shredder	3 kW

B. Size Reduction Section

Sr.	Machine	Function	Power
1	Double Shaft Shredder	Break batteries into 20–30 mm pieces	22 + 22 kW
2	Hammer Crusher	Further crushing	39.2 kW
3	Crusher Machine	Secondary crushing for fine release	39.2 kW
4	Grinding Machine	Powdering Cu/Al to release black mass	45 kW

Core crushing section (critical for black mass liberation)

C. Material Conveying System

Sr.	Machine	Function	Power
1	Wind Conveyor + Cyclone	Pneumatic conveying	7 kW
2	Cyclone Collector	Material collection	7 kW

Sr. Machine	Function	Power
3 Cyclone Collector (2 sets)	Feeding to screen	7 kW × 2
4 Cyclone Collector (2 sets)	Post-crushing handling	9 kW × 2
5 Cyclone Collector	Additional collection	12.5 kW
6 Bucket Elevator	Vertical lifting	3 kW

D. Primary Separation Section

Sr. Machine	Function
1 Zigzag Separator (shell removal)	Removes metal/plastic shells
2 Discharge Conveyor	Shell removal output
3 Drum Magnetic Separator	Removes iron

E. Secondary Separation Section

Sr. Machine	Function
1 Zigzag Separator (film removal) – 2 sets	Removes separator film
2 Round Vibrating Screen – 2 sets	Size separation
3 Black Powder Screening Machine – 2 sets	Fine black mass separation
4 Magnetic Separator – 2 sets	Removes residual metals

F. Final Metal Separation Section

Sr.	Machine	Function
1	Round Vibrating Screen	Final separation
2	Airflow Separator	Separates copper & aluminum

G. Storage & Collection

Sr.	Machine	Function
1	Centralized Collector Silo	Black mass storage

H. Dust & Pollution Control System

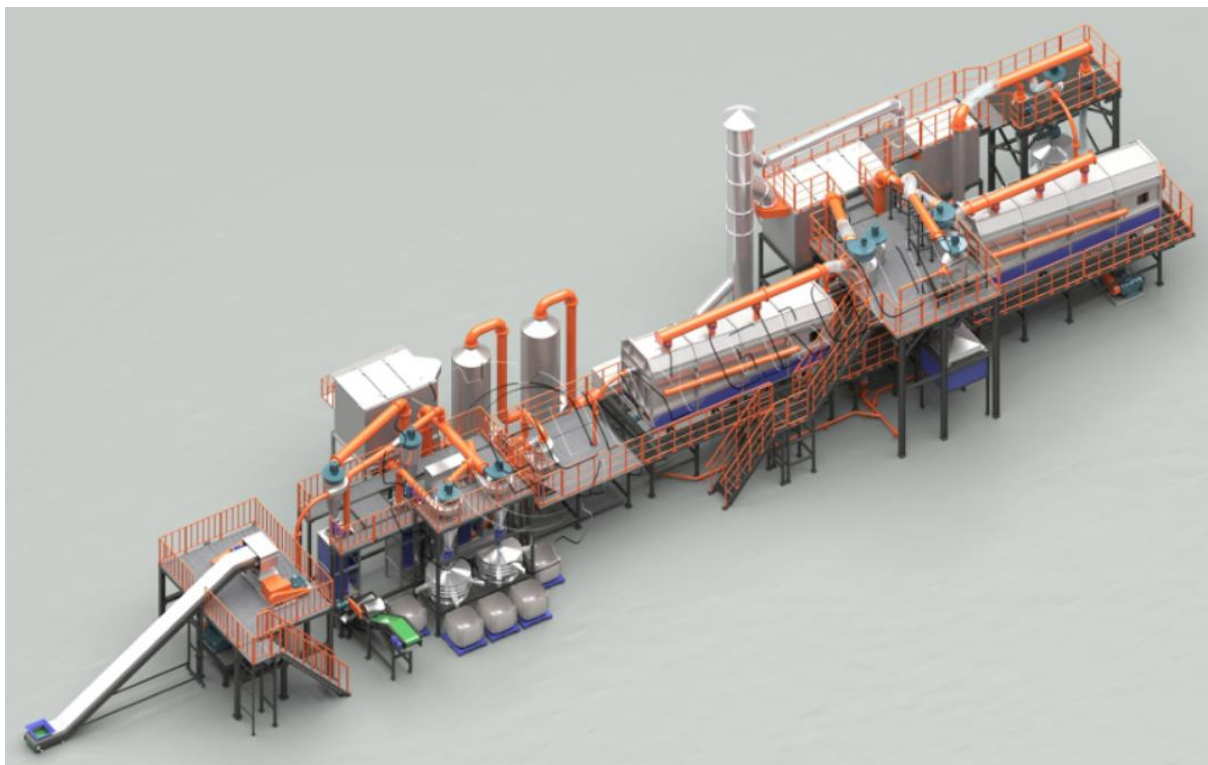
Sr.	Machine	Function
1	Pulse Dust Collector (96 bags)	Dust control
2	Pulse Dust Collector (200 bags)	Dust control
3	Pulse Dust Collector (150 bags)	Dust control
4	Spray Tower (2 sets)	Gas treatment
5	Activated Carbon Box	Odor removal
6	Industrial Fans	Air movement

I. Electrical & Supporting Systems

Sr.	Machine	Function
1	Walking Platform	Maintenance access
2	Electrical Control Panel PLC/control system	

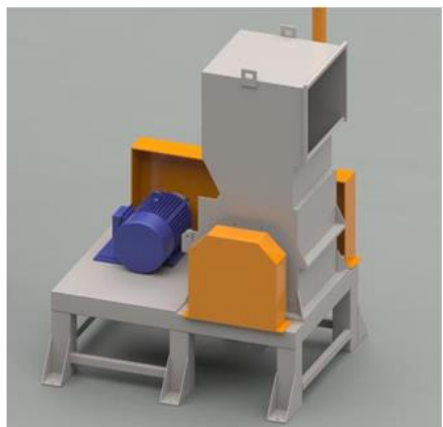
Process Flow

Feeding → Shredding → Crushing → Air Conveying
→ Zigzag Separation (shell removal)
→ Magnetic Separation
→ Film Separation
→ Screening
→ Grinding
→ Final Separation (Cu/Al/Black Mass)
→ Collection



Some important machineries photos

No.	Picture	Machine Name, Function & Specifications	
1		Feeding Belt Conveyor (with cover)	
		Z type with sealing cover Used to lift feeding the raw materials to the Shredder	
		Model	6-10-Z
		Power	3kw
		Belt size	10000*600mm

2		Double Shaft Shredder (on the platform) Shred batteries into 20-30mm pieces. The battery shells, anode/cathode sheets, separation films are turned into loose pieces <table><tr><td>Model</td><td>600 model</td></tr><tr><td>Power</td><td>22kw+22kw</td></tr><tr><td>Dimension</td><td>2500*1050*1850mm</td></tr></table>	Model	600 model	Power	22kw+22kw	Dimension	2500*1050*1850mm
Model	600 model							
Power	22kw+22kw							
Dimension	2500*1050*1850mm							
3		Hammer Crusher Crushing materials into smaller pieces <table><tr><td>Model</td><td>CP-600</td></tr><tr><td>Power</td><td>39.2kw</td></tr><tr><td>Dimension</td><td>2000*1400*2200mm</td></tr></table>	Model	CP-600	Power	39.2kw	Dimension	2000*1400*2200mm
Model	CP-600							
Power	39.2kw							
Dimension	2000*1400*2200mm							
4		Wind Conveyor with Cyclone Collector Carry the materials to next step There are wind conveyors during the whole process <table><tr><td>Model</td><td>600 model</td></tr><tr><td>Power</td><td>7kw</td></tr></table>	Model	600 model	Power	7kw		
Model	600 model							
Power	7kw							
5		Zigzag Separator for removing shells Used to separate the battery shells Heavy materials (iron, aluminum, plastic shells) drop down to the discharge conveyor Light materials (anode&cathode pieces, separator film, black mass) go to next step <table><tr><td>Model</td><td>700</td></tr></table>	Model	700				
Model	700							

		Size	1200*1200*3200mm
6		Discharge Belt Conveyor for Shells	
		Used to discharge iron, aluminum and plastic shells	
		Model	500-5-Z
		Power	3kw
		Belt size	5000*500mm
7		Drum Type Magnetic Separator	
		Used to separate the iron shells from mixture shells Hanging over the discharge conveyor	
		Power	1.5kw
		Magnetic force	3000Gauss
9		Zigzag Separator for removing separator film*2 sets	
		Used to remove the plastic separator film Heavy materials (black mass) drop down and be conveyed to the Round vibrating screen Light materials (anode&cathode pieces, separator film) go to further processing	
		Model	700
		Size	1200*1200*3200mm
	Round Vibrating Screen*2 sets		
	2 screen mesh, 3 layers, 3 outlets. There is a small amount of plastic film in the mixture. Separate by the sizes, and improve the purity of black mass		
	Model	1500	
	Power	2.2kw*2 sets	
	Size	1500*1500mm	
	Black Powder Screening Machine*2 sets		
	Rolling screen to remove the black powder		
	Model	1200	
	Power	8.5kw*2 sets	

	Crusher Machine	
	Crush the mixture (anode&cathode pieces, separator film) into smaller powder/granules, and make the black mass fall off	
	Model	600
	Power	39.2kw
	Centralized Collector Silo (optional)	
	There are pipes to convey the black powder from different outlets to this one silo. Easier to collect them	
	Power	1.5kw
	Grinding Machine	
	Used to grind the aluminum and copper granules into powder, to release the black mass and easier for further separation	
	Model	600
	Power	45kw
	Size	2200*1900*4200mm
Fan*2 sets		
	Round Vibrating Screen	
	3 screen mesh, 4 layers, 4 outlets. Used to separate the metal, plastic and black mass	
	Model	1200
	Power	1.1kw

	Airflow Separator	
	Separate the copper and aluminum	
	Model	600
	Power	6kw
	Size	1200*1500*1500
	Pulse Dust Collector 1	
	Used to collect the dust during the process	
	Model	96 bags
	Power	1.5kw
	Pulse Dust Collector 2	
	Used to collect the dust during the process	
	Model	200 bags
	Power	3kw
	Pulse Dust Collector 3	
	Used to collect the dust during the process	
	Model	150 bags
	Power	3kw
	Spray Tower*2 sets	
	Used to get rid of the exhaust gas	
	Power	2.2kw*2 sets

Hydrometallurgical part

PART 1: Black Mass Handling & Metal Separation (Mechanical + Pre-treatment)

A. Black Mass Processing & Pre-treatment Machinery

Sr. No.	Machinery	Description / Function
1	Black Mass Storage System	Storage silos / bins for incoming black mass
2	Black Mass Conveying System	Screw / belt conveyors for feeding
3	Rotary Furnace & Allied System	Thermal treatment (removal of organics, binder)
4	Bag Filter System	Dust collection during handling
5	Carbon Column	Removal of impurities / organic residues
6	Tanks & Settlers	Slurry preparation before leaching
7	Pumps with Motors	Slurry transfer
8	Filter Presses	Solid-liquid separation

These systems prepare **black mass for hydrometallurgical processing** by:

- Removing impurities
- Stabilizing material
- Converting into slurry form

PART 2: Hydrometallurgy Plant Machinery

B. Leaching & Purification Section

Sr. No.	Machinery	Function
1	SS316L Leaching Reactors with Agitator	Acid leaching of metals
2	Tanks & Settlers	Solution holding & separation
3	Pumps with Motors	Chemical/slurry transfer
4	Product Purification & Precipitation Reactors	Removal of impurities
5	Bag Filter	Fine filtration
6	Carbon Column	Organic impurity removal

C. Solvent Extraction (SX) System

Sr. No.	Machinery	Function
1	Mixer-Settlers (Cu, Co, Ni, Mn)	Selective metal separation
2	Organic Storage Tanks	Solvent storage
3	Aqueous Tanks	Process solution holding
4	Pumps & Piping	Fluid circulation
5	Utilities Integration	Flow control system

This is the **most critical value extraction stage**

D. Evaporation & Crystallization Section

Sr. No.	Machinery	Function
1	Evaporation Plant (Manganese Sulphate)	Concentration of Mn solution
2	Evaporation Plant (Sodium/Lithium Sulphate)	Lithium recovery concentration
3	Crystallizers	Salt formation
4	Centrifuge	Solid-liquid separation
5	Dryers	Final product drying

E. Electro winning Section (Metal Recovery)

Cobalt Electro winning

Machinery

Rectifier

EW Cells

Lead Anodes

Cathodes

Pumps with Motors

Nickel Electro winning

Machinery

Rectifier

EW Cells

Lead Anodes

Cathodes

Heat Exchangers

Tanks & Filters

F. Product Handling & Finishing

Sr. No.	Machinery	Function
1	Cathode Washing & Stripping Machine	Metal recovery from cathodes
2	Product Handling System	Packing & storage
3	Cranes & Material Handling Equipment	Internal movement
4	Busbars	Electrical conduction in EW

G. Utilities Section

Sr. No.	Machinery	Function
1	Boiler	Steam generation
2	Cooling Tower	Cooling water
3	DM Plant	Pure water supply
4	Storage Tanks	Utility storage

H. Instrumentation & Electricals

Sr. No.	Machinery	Function
1	Instrumentation System (~5%)	Process control
2	Electrical Panels (~10%)	Power distribution
3	Piping (~5%)	Fluid transfer

Your process is designed to produce:

- Cobalt → 99.95% purity
- Lithium Carbonate → 99.9%
- Nickel Sulphate → 99.95%
- Manganese Sulphate → 99.6%

The plant will include black mass handling and treatment systems comprising storage, conveying, Rotary furnace, filtration, and slurry preparation units. The hydrometallurgical section will consist of leaching reactors, purification systems, solvent extraction units, evaporation and crystallization systems, and electro winning units for recovery of cobalt and nickel metals. The facility will also include utilities such as boiler, cooling systems, DM water plant, and integrated instrumentation for process control.

Final Recommendation for a Premium Lithium-Ion Battery Recycling Plant

Tier I – Premium Global Standard

1. Genox Recycling Technology Co., Ltd.

Best for: Investors prioritizing maximum equipment reliability, premium engineering, long service life and excellent global after-sales support.

2. Techcycle Technology Co., Ltd.

Best for: Integrated recycling projects with a long-term vision of expanding into hydrometallurgical recovery and critical mineral production.

Tier II – High-Quality Commercial Solutions

3. DOING Holdings

Best for: Commercial projects seeking a strong balance between cost, performance and turnkey execution.

4. Guanma Machinery

Best for: Medium-scale premium projects where competitive pricing is desired without significant compromise on product quality or service.

Tier III – Reliable Mid-Premium Solution

5. Kerui Machinery

Best for: Investors requiring dependable equipment with good recovery efficiency and moderate capital investment.

13. Process flow

PROCESS FLOW AND MASS BALANCE

5 TPD Lithium-Ion Battery Recycling Plant with 0.5 TPD Hydrometallurgical Processing

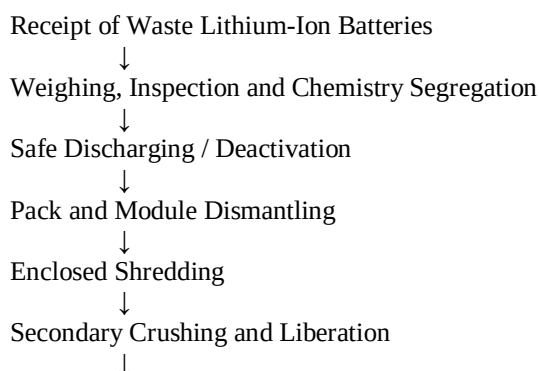
1. Project Configuration

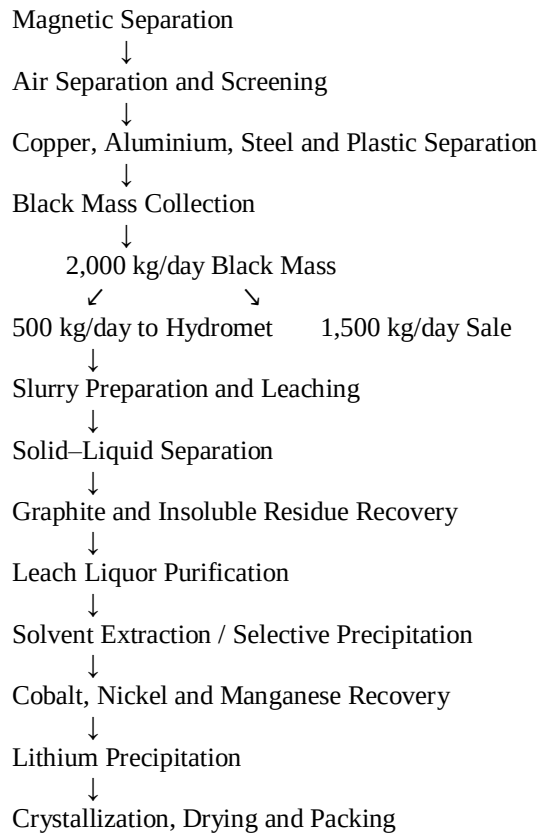
Parameter	Design Basis
Waste lithium-ion battery processing capacity	5,000 kg/day
Black mass recovery from battery scrap	40%
Black mass generation	2,000 kg/day
Hydrometallurgical processing capacity	500 kg/day
Surplus black mass available for sale	1,500 kg/day
Operating days	300 days/year
Annual battery scrap processing	1,500 MT/year
Annual hydrometallurgical feed	150 MT/year

The facility will operate through two integrated stages:

1. **Mechanical recycling and black mass production**
2. **Hydrometallurgical recovery of metals from part of the black mass**

2. Overall Process Flow





3. Stage 1: Battery Receipt and Segregation

The incoming batteries will be inspected, weighed and segregated according to battery chemistry and physical condition.

Battery types may include:

- NMC batteries
- LCO batteries
- NCA batteries
- LFP batteries
- Mixed consumer-electronics batteries
- EV battery modules and cells
- Battery manufacturing rejects

Damaged, swollen or thermally unstable batteries will be isolated in a designated fire-safe quarantine area before processing.

4. Stage 2: Discharging and Dismantling

Battery packs and modules will be electrically discharged to reduce fire and electrical hazards. Large battery packs will be manually or mechanically dismantled to remove:

- Outer casing
- Busbars
- Wiring
- Battery management systems
- Cooling plates
- Structural steel and aluminium components

The cells and smaller battery fractions will then be transferred to the enclosed shredding system.

5. Stage 3: Mechanical Recycling

The mechanical processing section will include:

1. Feeding
2. Primary shredding
3. Secondary crushing
4. Size classification
5. Magnetic separation
6. Air classification
7. Plastic and separator removal
8. Copper and aluminium separation
9. Black mass collection

The processing line will operate under negative pressure with cyclone separators, bag filters and other dust-control systems.

6. Mechanical Recycling Mass Balance

Input: 5,000 kg/day Battery Scrap

Output Stream	Recovery Basis	Quantity
Black mass	40%	2,000 kg/day
Copper, aluminium and steel fractions	25%	1,250 kg/day
Plastics and separator material	20%	1,000 kg/day
Moisture, electrolyte and process losses	15%	750 kg/day
Total	100%	5,000 kg/day

Annual Mechanical Output

Output Stream	Annual Quantity at 300 Days
Black mass	600 MT/year
Copper, aluminium and steel fractions	375 MT/year
Plastics and separators	300 MT/year
Moisture, electrolyte and process losses	225 MT/year
Total	1,500 MT/year

The 750 kg/day stream includes electrolyte vapour captured through the emission-control system, moisture, fine process residue, contaminated separator fractions and unavoidable handling losses. It should not be treated entirely as atmospheric loss.

7. Black Mass Allocation

Black Mass Stream	Quantity
Total black mass generated	2,000 kg/day
Black mass processed through hydrometallurgy	500 kg/day
Black mass available for external sale	1,500 kg/day

Annual Allocation

Black Mass Stream	Annual Quantity
Hydrometallurgical feed	150 MT/year
Black mass sale	450 MT/year
Total black mass	600 MT/year

8. Indicative Black Mass Composition

The actual composition will vary according to battery chemistry. For design and financial evaluation, the following mixed NMC/LCO-rich black mass composition has been considered.

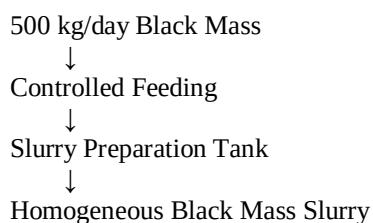
Component	Composition	Quantity in 500 kg/day Feed
Cobalt	25%	125 kg/day
Nickel	15%	75 kg/day
Manganese	10%	50 kg/day
Lithium	4%	20 kg/day
Graphite and carbon	30%	150 kg/day
Binder, aluminium fines and impurities	16%	80 kg/day
Total	100%	500 kg/day

This composition must be confirmed through laboratory analysis before finalizing reactor size, reagent consumption and product-recovery estimates.

9. Hydrometallurgical Process Flow

9.1 Black Mass Feeding and Slurry Preparation

The black mass will be transferred from a closed storage silo into a slurry preparation tank. Process water and controlled quantities of reagents will be added to create a uniform slurry.



9.2 Acid Leaching

The slurry will be treated in agitated reactors using a suitable leaching system, generally based on:

- Sulphuric acid
- Hydrogen peroxide or another reducing agent
- Process water
- Controlled temperature and pH

During leaching, cobalt, nickel, manganese and lithium dissolve into the leach solution. Graphite and certain impurities remain in the solid phase.

Black Mass Slurry

↓
Acid and Reducing Agent Addition
↓
Controlled Leaching
↓
Metal-Rich Leach Slurry

9.3 Solid–Liquid Separation

The leached slurry will be passed through a filter press or equivalent filtration system.

Outputs:

- Metal-rich pregnant leach solution
- Graphite-rich filter cake
- Insoluble impurities

The filter cake may be washed to recover entrained metals and improve graphite purity.

9.4 Leach Liquor Purification

The metal-rich solution will undergo purification to remove:

- Iron
- Aluminium
- Copper traces
- Fluoride and phosphate impurities
- Other unwanted dissolved contaminants

Purification will generally involve:

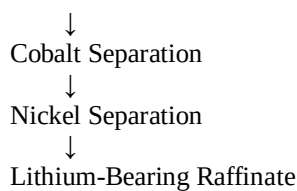
- Controlled pH adjustment
- Chemical precipitation
- Filtration
- Carbon treatment, where required

9.5 Solvent Extraction and Selective Metal Recovery

The purified leach liquor will be processed through solvent extraction or selective precipitation for separation of cobalt, nickel and manganese.

Typical sequence:

Purified Leach Liquor
↓
Copper and Impurity Removal
↓
Manganese Separation



The precise sequence may change depending on product specifications and the selected technology provider.

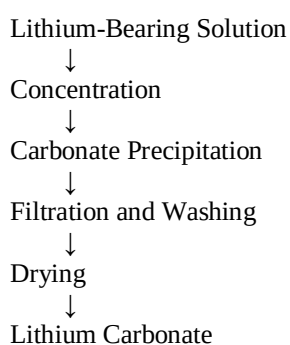
9.6 Product Recovery

The separated metals may be recovered in the following forms:

Metal	Possible Final Product
Cobalt	Cobalt sulphate, cobalt hydroxide or cobalt metal equivalent
Nickel	Nickel sulphate, nickel hydroxide or nickel metal equivalent
Manganese	Manganese sulphate monohydrate
Lithium	Lithium carbonate
Graphite	Recovered graphite concentrate

9.7 Lithium Recovery

After removal of cobalt, nickel and manganese, the lithium-bearing solution will be concentrated and treated with a carbonate reagent to precipitate lithium carbonate.



10. Metal Recovery Assumptions

Component Recovery Efficiency

Cobalt	95%
--------	-----

Component Recovery Efficiency

Nickel	95%
Manganese	95%
Lithium	90%
Graphite	95%

Graphite recovery has been considered at 95% for project planning. A 100% recovery assumption is generally unsuitable for a bankable mass balance because filtration, washing and handling losses are unavoidable.

11. Hydrometallurgical Metal Balance

Component	Input	Recovery	Recovered Metal Equivalent	Loss to Residue
Cobalt	125 kg/day	95%	118.75 kg/day	6.25 kg/day
Nickel	75 kg/day	95%	71.25 kg/day	3.75 kg/day
Manganese	50 kg/day	95%	47.50 kg/day	2.50 kg/day
Lithium	20 kg/day	90%	18.00 kg/day	2.00 kg/day
Graphite	150 kg/day	95%	142.50 kg/day	7.50 kg/day
Impurities and binder	80 kg/day	—	—	80.00 kg/day
Total	500 kg/day		398.00 kg/day	102.00 kg/day

The 398 kg/day recovered quantity is presented on an elemental or graphite-equivalent basis.

12. Final Product Balance

12.1 Products on Metal-Equivalent Basis

Product	Output
Cobalt product, metal equivalent	118.75 kg/day
Nickel product, metal equivalent	71.25 kg/day
Manganese, metal equivalent	47.50 kg/day

Product	Output
Lithium, metal equivalent	18.00 kg/day
Recovered graphite	142.50 kg/day
Process residue and losses	102.00 kg/day
Total	500.00 kg/day

12.2 Products on Saleable Compound Basis

Where manganese is recovered as manganese sulphate monohydrate and lithium is recovered as lithium carbonate, the indicative product quantities will be:

Final Product	Indicative Output
Cobalt product, cobalt equivalent	118.75 kg/day
Nickel product, nickel equivalent	71.25 kg/day
Manganese sulphate monohydrate	Approximately 146 kg/day
Lithium carbonate	Approximately 96 kg/day
Recovered graphite	Approximately 142.5 kg/day
Process residue and sludge	Approximately 102 kg/day

The combined product mass on compound basis is higher than the elemental metal mass because sulphate, carbonate, oxygen and water-of-crystallization are introduced through process reagents.

Therefore, saleable compound output should not be directly added to the original black mass feed without separately accounting for reagent-derived mass.

13. Annual Hydrometallurgical Product Output

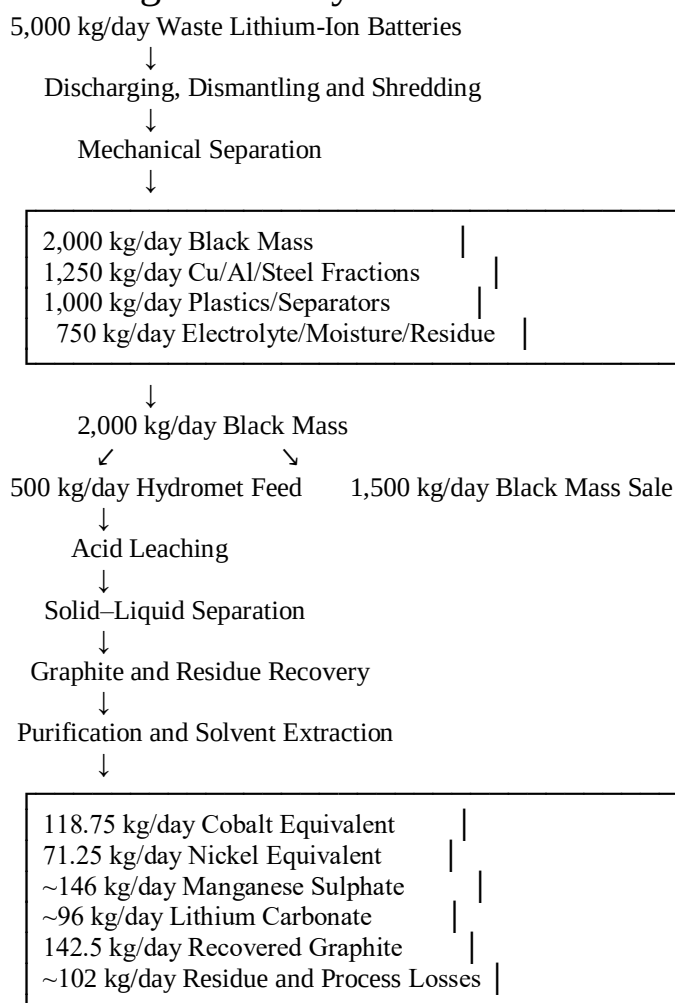
At 300 operating days per year:

Product	Annual Output
Cobalt product, metal equivalent	35.63 MT/year
Nickel product, metal equivalent	21.38 MT/year
Manganese sulphate monohydrate	Approximately 43.8 MT/year

Product	Annual Output
Lithium carbonate	Approximately 28.8 MT/year
Recovered graphite	Approximately 42.75 MT/year
Process residue and sludge	Approximately 30.6 MT/year

These figures are indicative and will depend on actual feed chemistry, moisture, product purity and operating recovery.

14. Integrated Daily Mass Flow



15. Process Residue and Waste Handling

The hydrometallurgical process will generate:

- Graphite-rich filter cake
- Iron and aluminium hydroxide sludge
- Metal-bearing filter residue
- Spent filter media

- ETP sludge
- Evaporation salts, where ZLD is adopted
- Used solvent and contaminated process consumables

Recoverable graphite will be processed as a saleable material where purity permits. Remaining hazardous residues will be stored in a dedicated hazardous-waste storage area and transferred to an authorized recycler, co-processor or TSDF in accordance with the applicable authorization.

16. Water and Solution Management

The hydrometallurgical process will use process water for:

- Slurry preparation
- Leaching
- Filter cake washing
- Product washing
- Chemical solution preparation
- Scrubber operation
- Floor and equipment cleaning

Process filtrate, wash water and condensate should be recycled as far as technically feasible. The facility should operate with an ETP and, where required by the Pollution Control Board, a Zero Liquid Discharge system.

A separate detailed water balance will be finalized after selection of:

- Liquid-to-solid ratio
- Leaching chemistry
- Number of washing stages
- Evaporation system
- Product purity requirement

17. Key Design Considerations

1. Battery chemistry must be segregated before mechanical processing.
2. LFP-rich and LTO-rich feed should not be mixed indiscriminately with NMC/LCO feed.
3. Black mass composition must be verified through accredited laboratory testing.
4. Chemical consumption and product yield must be based on assay results.
5. The mechanical line must remain fully enclosed and under negative pressure.
6. Hydromet reactors, tanks and piping must be compatible with acidic and oxidizing chemicals.
7. Solvent extraction areas require bunding, ventilation and fire protection.
8. Separate storage must be provided for acids, alkalis, solvents and recovered products.

9. Process-water recycling and metal-bearing wastewater treatment are essential.
10. Product quantities must clearly distinguish between metal-equivalent basis and chemical-compound basis.

18. Conclusion

The proposed integrated facility will process **5,000 kg/day of lithium-ion battery waste**, producing approximately **2,000 kg/day of black mass** through mechanical separation. Of this quantity, **500 kg/day will be processed internally through hydrometallurgy**, while approximately **1,500 kg/day may be sold as black mass**.

Based on the assumed mixed black mass composition and recovery efficiencies, the hydrometallurgical section can recover approximately:

- **118.75 kg/day cobalt equivalent**
- **71.25 kg/day nickel equivalent**
- **146 kg/day manganese sulphate monohydrate**
- **96 kg/day lithium carbonate**
- **142.5 kg/day recovered graphite**

The mass balance is indicative and suitable for preliminary project planning. Final machinery design, chemical consumption, revenue estimation and product output must be based on representative feed analysis and technology-provider performance guarantees.

14. Layout



14. ETP and ECS requirement

ETP AND EMISSION CONTROL SYSTEM REQUIREMENTS

5 TPD Lithium-Ion Battery Recycling Plant with 0.5 TPD Hydrometallurgical Processing

1. Design Basis

The proposed facility comprises:

Section	Capacity
Mechanical battery recycling	5,000 kg/day
Black mass generation	Approximately 2,000 kg/day
Hydrometallurgical processing	500 kg/day black mass
Surplus black mass sale	Approximately 1,500 kg/day

The mechanical section will primarily generate dust, electrolyte vapours and fine metallic particles, while the hydrometallurgical section will generate acidic fumes, metal-bearing wastewater, wash water, filter cake and chemical sludge.

The plant must operate with effective pollution-control systems and environmentally sound recycling infrastructure in accordance with the Battery Waste Management Rules, 2022, subsequent amendments, and conditions imposed by the concerned State Pollution Control Board.

2. Effluent Treatment Plant Requirement

2.1 Sources of Wastewater

The principal wastewater streams will arise from the 0.5 TPD hydrometallurgical section.

Source	Nature of Wastewater
Reactor and slurry tank washing	Acidic water containing dissolved metals
Filter press and filter-cake washing	Metal-bearing wash water
Product washing	Sulphate, carbonate and residual metal content
Floor and equipment washing	Suspended solids, metals and chemicals

Source	Nature of Wastewater
Wet scrubber blowdown	Acidic/alkaline water with dissolved pollutants
Laboratory wastewater	Acids, alkalis and trace metals
Cooling-tower blowdown	High TDS
RO/DM plant reject	High TDS
Domestic use	Sewage

No process wastewater is expected from a properly designed dry mechanical-separation line, except floor washing, scrubber blowdown and accidental spill collection.

2.2 Estimated Effluent Quantity

For preliminary design, the following water and wastewater generation may be considered:

Stream	Estimated Quantity
Hydromet process and washing wastewater	2.0–3.0 KLD
Scrubber blowdown	0.5–1.0 KLD
Floor/equipment washing	0.5 KLD
Laboratory and miscellaneous wastewater	0.2 KLD
RO/DM reject and utility blowdown	0.5–1.0 KLD
Total industrial effluent	Approximately 4–5 KLD

The ETP should therefore be designed for:

Recommended ETP Capacity: 7.5 KLD

A 7.5 KLD system provides approximately 50% hydraulic margin for variation in washing frequency, product purification, scrubber blowdown and future operational expansion.

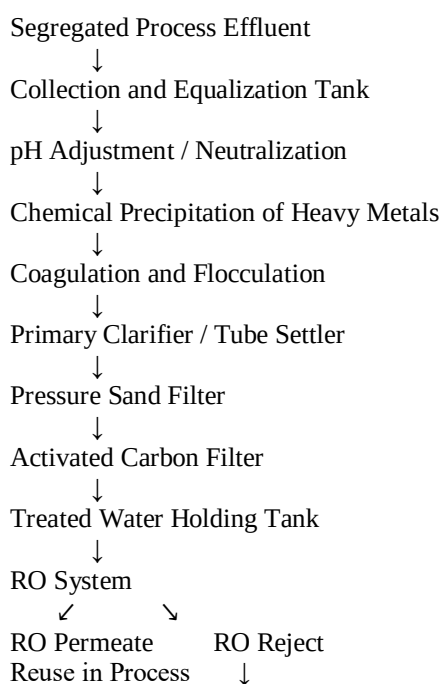
Where the hydromet section is later expanded beyond 0.5 TPD, the ETP should be modularly expandable to at least **15 KLD**.

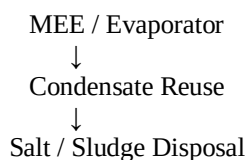
3. Expected Effluent Characteristics

Parameter	Expected Nature
pH	Strongly acidic to alkaline, depending on batch
TDS	High
Sulphates	High
Suspended solids	Moderate to high
COD	Low to moderate
Cobalt	Present
Nickel	Present
Manganese	Present
Lithium	Present
Iron and aluminium	Present
Fluoride/phosphate	Possible, depending on battery chemistry
Solvent traces	Possible near SX section

The treatment plant must be designed around actual wastewater characterization obtained after technology and reagent finalization.

4. Recommended ETP Process Flow





5. ETP Equipment Requirement

Equipment	Recommended Capacity
Process-effluent collection tank	10 KL
Equalization tank	10 KL
Emergency spill-holding tank	5 KL
Neutralization tank	5 KL
Heavy-metal precipitation tank	5 KL
Coagulation/flocculation tank	3–5 KL
Tube settler/clarifier	7.5 KLD
Filter feed tank	5 KL
Pressure sand filter	1–2 m ³ /hour
Activated carbon filter	1–2 m ³ /hour
Treated-water tank	10 KL
Filter press	Suitable for 50–100 kg/day wet sludge
RO plant	5 KLD
Multiple-effect evaporator or suitable evaporator	1–2 KLD reject capacity
Sludge-storage area	15–20 m ²
Chemical-dosing system	Acid, alkali, coagulant and flocculant

6. Treatment Chemistry

The ETP will typically require:

- Lime or sodium hydroxide for neutralization;
- Ferrous/ferric salts or appropriate coagulants;
- Polyelectrolyte for flocculation;
- Selective metal-precipitation chemicals;
- Sulphide-based precipitation, only where technically required and safely controlled;
- Activated carbon for organic and solvent traces;

- pH and oxidation-reduction potential monitoring.

Metal-rich streams should first be evaluated for internal recovery before they are diverted to the ETP. Directly sending valuable cobalt-, nickel- or lithium-bearing solutions to the ETP will increase chemical consumption and metal loss.

7. Zero Liquid Discharge Requirement

For a hydrometallurgical battery-recycling facility, the preferred design basis is:

Zero Liquid Discharge or Maximum Water Recycle

The plant should not discharge untreated or partially treated process wastewater into:

- Open drains;
- Storm-water drains;
- Land;
- Sewer without authorization;
- Surface-water bodies.

The final ZLD requirement will be specified through the Consent to Establish and Consent to Operate conditions of the relevant State Pollution Control Board. The project should be designed for internal reuse of RO permeate, evaporator condensate and treated scrubber water.

The regulatory framework requires environmentally sound recycling and compliance with applicable effluent, emission and hazardous-waste conditions. (Central Pollution Control Board)

8. Sewage Treatment Requirement

Domestic sewage should remain separate from industrial effluent.

Assuming 25–35 employees:

Parameter	Indicative Value
Domestic water consumption	1.5–2.0 KLD
Sewage generation	1.2–1.6 KLD
Recommended treatment	Septic tank/soak pit where permitted, or compact STP
Recommended STP capacity	3 KLD

Treated domestic wastewater may be used for greenbelt development and floor washing, subject to consent conditions.

9. Emission Control System Requirement

The Emission Control System must separately address:

1. Dust and black-mass particles;
2. Electrolyte vapours and VOCs;
3. Acid fumes from hydrometallurgy;
4. Boiler or thermic-fluid-heater emissions;
5. Fugitive emissions from storage and material handling.

10. Mechanical Recycling Section — Emission Sources

Process	Likely Emission
Battery opening and dismantling	Dust and electrolyte vapours
Shredding	Black-mass dust, metal fines and VOCs
Crushing and grinding	Fine particulate matter
Screening	Black-mass dust
Air separation	Fine particles
Conveyor transfer points	Fugitive dust
Black-mass packing	Fine metal-oxide and graphite dust

Black mass contains very fine particles and must be handled as an enclosed powder under negative pressure.

11. Recommended APCS for Mechanical Section

Enclosed Shredder / Crusher / Screens



Local Exhaust Ventilation Hoods



Common Negative-Pressure Ducting



Spark Arrestor / Drop-out Chamber



Cyclone Separator



Pulse-Jet Bag Filter



Activated Carbon Adsorption Unit
 ↓
 ID Fan
 ↓
 Common Stack

A wet scrubber should not replace the bag filter for black-mass dust. The bag filter is the principal particulate-control device, while activated carbon or a polishing scrubber may be used for electrolyte vapours and odour.

12. Mechanical-Section ECS Capacity

For a 5 TPD line, the final airflow must be based on hood dimensions, number of extraction points and equipment vendor data. The following preliminary capacities may be adopted:

System	Recommended Design Capacity
Primary shredder and crusher extraction	8,000–10,000 m ³ /hour
Screening and separation extraction	6,000–8,000 m ³ /hour
Black-mass collection and packing	3,000–5,000 m ³ /hour
Total combined design airflow	Approximately 18,000–25,000 m ³ /hour
Cyclone separator	20,000–25,000 m ³ /hour
Pulse-jet bag filter	25,000 m ³ /hour
ID fan	25,000–30,000 m ³ /hour
Activated-carbon polishing unit	Sized for full outlet airflow

For improved reliability, the mechanical section may use two separate bag-filter systems instead of one common unit:

- **System A:** shredder, crusher and grinder;
- **System B:** screens, separators, collection and packing.

13. Bag Filter Design Criteria

Parameter	Recommended Requirement
Filter type	Pulse-jet bag filter
Filter media	Antistatic, chemical-resistant media
Air-to-cloth ratio	Approximately 0.8–1.2 m/min

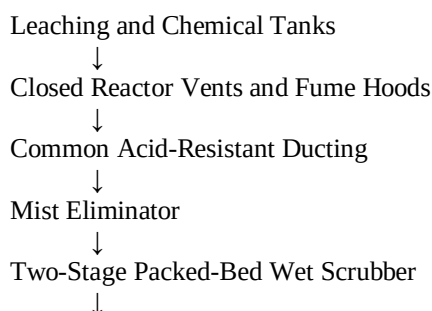
Parameter	Recommended Requirement
Dust handling	Enclosed screw conveyor or sealed drum
Differential-pressure gauge	Mandatory
Spark protection	Required upstream
Explosion venting/isolation	To be evaluated
Dust return	Recycle to black mass where quality permits
Outlet PM design target	Preferably below 30 mg/Nm ³ or consent limit, whichever is stricter

The dust collector should be fitted with antistatic bags, earthing, explosion isolation and automatic pulse cleaning because fine graphite and metal powder may present combustible-dust risks.

14. Hydrometallurgy Section — Emission Sources

Process	Likely Emission
Acid storage and dosing	Acid vapour
Leaching reactors	Acid mist and process fumes
Neutralization	Heat and chemical vapour
Solvent extraction	Organic vapours
Precipitation	Acidic or alkaline mist
Evaporation/crystallization	Vapour and entrained droplets
Product drying	Product dust
Laboratory operations	Chemical fumes

15. Recommended Scrubber System for Hydrometallurgy



Demister
↓
ID Fan
↓
Dedicated Stack

Recommended Capacity

System	Design Capacity
Leaching and reaction-area extraction	5,000–7,500 m ³ /hour
Chemical-storage and dosing extraction	1,500–2,500 m ³ /hour
Laboratory fume extraction	1,000–1,500 m ³ /hour
Total scrubber capacity	8,000–10,000 m ³ /hour

A **10,000 m³/hour two-stage packed-bed scrubber** is recommended for the initial 0.5 TPD hydromet plant, with sufficient space for installation of a second parallel scrubber during expansion.

16. Scrubber Design Requirements

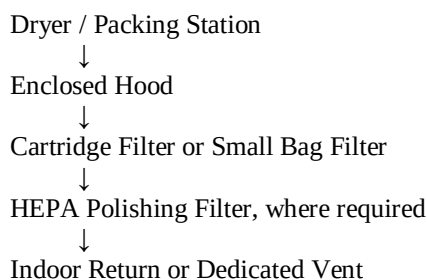
Component	Requirement
Construction material	FRP/PP/PVC or compatible corrosion-resistant material
Scrubber type	Two-stage packed-bed scrubber
Stage 1	Acid-fume neutralization
Stage 2	Polishing/alkaline scrubbing
Recirculation tank	3–5 KL
Recirculation pumps	One working plus one standby
pH monitoring	Online
Pressure-drop indicator	Required
Mist eliminator	Required
Blowdown connection	To ETP
Stack	Dedicated, as approved by SPCB

Solvent-extraction vapours should not be routed through an unsuitable common acid scrubber without compatibility assessment. Where solvent vapours are significant, a dedicated activated-carbon adsorption system and flameproof ventilation should be provided.

17. Product-Drying Dust Control

Drying and packing of recovered salts and graphite may generate fine product dust.

Recommended system:



Parameter	Recommended Capacity
Graphite/product dryer extraction	2,000–3,000 m ³ /hour
Product-packing extraction	1,000–2,000 m ³ /hour
Filter type	Cartridge or fine bag filter
Dust collection	Sealed, product-compatible container

18. Boiler or Heating-System Emissions

Where a boiler or thermic-fluid heater is installed, clean fuel should be preferred. Preferred Fuels

1. PNG;
2. LPG;
3. Electricity;
4. Approved low-emission fuel.

Coal, pet coke and high-sulphur liquid fuels should be avoided because they increase pollution-control requirements and may complicate approval, particularly in Haryana/NCR-influenced locations.

The heating-unit stack height and emission standards will be determined according to:

- Fuel type;
- Heat capacity;
- Boiler rating;
- State Pollution Control Board conditions;
- Applicable CPCB standards.

19. Stack Requirement

Separate stacks are recommended for:

Stack	Connected System
Stack 1	Mechanical dust-control bag filter
Stack 2	Hydromet wet scrubber
Stack 3	Boiler/thermic-fluid heater
Stack 4	DG set, where installed

Indicative stack heights may be considered as follows, subject to SPCB approval:

System	Indicative Height
Mechanical APCS	15–20 m
Hydromet scrubber	15–20 m
Boiler	As per fuel and capacity formula/consent
DG set	Building height plus prescribed additional height

All process stacks should have:

- Sampling porthole;
- Monitoring platform;
- Safe ladder or staircase;
- Handrail;
- Electrical socket;
- Adequate straight duct length;
- Stack identification plate.

20. Fugitive Emission Control

The following measures are required:

- Fully enclosed conveyors and transfer points;
- Negative-pressure operation in shredding and black-mass zones;
- Sealed black-mass silos and packing stations;
- Vacuum cleaning instead of dry sweeping;
- Antistatic industrial flooring;

- Double-door entry in black-mass areas;
- Separate PPE change room;
- Local exhaust ventilation at chemical dosing points;
- Covered acid and solvent storage;
- Bund walls around chemical tanks;
- Regular leak inspection;
- No open handling of black mass;
- Continuous housekeeping and spill management.

21. Monitoring and Instrumentation

ETP Monitoring

- Online pH;
- Flow meter;
- TDS/conductivity;
- Metal testing for cobalt, nickel, manganese, lithium, iron and aluminium;
- Daily chemical-dosing records;
- Sludge-generation records;
- Treated-water reuse records.

ECS Monitoring

- Bag-filter differential pressure;
- Scrubber pH;
- Scrubber circulation flow;
- Fan current and airflow;
- Stack particulate monitoring;
- Acid-mist monitoring where prescribed;
- Workplace dust monitoring;
- VOC monitoring near shredding and SX areas;
- Records of bag replacement and scrubber blowdown.

22. Hazardous Waste from ETP and ECS

Waste Stream	Disposal Route
ETP heavy-metal sludge	Authorized TSDF or authorized recovery facility
Bag-filter dust	Recycle to black mass where technically suitable
Spent activated carbon	Authorized hazardous-waste facility
Scrubber sludge	ETP treatment and authorized disposal
Evaporator salts	Characterization followed by authorized disposal

Waste Stream	Disposal Route
Used filter bags	Sealed storage and authorized disposal
Spent solvents	Authorized recycler/recovery facility
Used oil	Authorized used-oil recycler

Hazardous waste must be handled under the applicable authorization, with proper labelling, storage, manifest documentation and disposal records.

23. Recommended Area Allocation

Facility	Recommended Area
ETP and ZLD section	150–200 m ²
Hydromet scrubber area	50–75 m ²
Mechanical bag-filter area	100–150 m ²
Hazardous-waste storage	75–100 m ²
Chemical storage and bunding	100–150 m ²
Emergency spill tank and utility area	50–75 m ²
Future ETP/ECS expansion	At least 150 m ² reserved

The pollution-control area should be located downwind and toward the rear or utility side of the plot, while remaining accessible to tankers, maintenance teams and hazardous-waste vehicles.

24. Recommended Final Configuration

ETP

- 7.5 KLD physico-chemical ETP;
- 5 KLD RO plant;
- 1–2 KLD evaporator for concentrated reject;
- 5 KL emergency spill tank;
- Filter press;
- Complete treated-water reuse;
- Reserved area for expansion to 15 KLD.

Mechanical ECS

- Local exhaust ventilation at all dust points;
- Cyclone pre-separator;
- Two pulse-jet bag-filter systems with combined capacity of approximately 25,000 m³/hour;
- Activated-carbon polishing;
- Negative-pressure production area;
- Dedicated process stack.

Hydromet ECS

- Closed reactors;
- FRP/PP corrosion-resistant ducting;
- 10,000 m³/hour two-stage packed-bed scrubber;
- Demister;
- Activated-carbon system for solvent vapours where required;
- Dedicated stack.

15. CAPEX, OPEX and Utility Requirements

CAPEX, OPEX, UTILITY REQUIREMENT AND FINANCIAL PROJECTIONS

500 kg/hour Mechanical Battery Recycling Plant with 500 kg/day Hydrometallurgy Plant

1. Project Configuration

Particular	Capacity
Mechanical recycling line	500 kg/hour
Battery scrap processing capacity	5,000 kg/day
Black mass generation assumption	40% of battery input
Black mass produced	2,000 kg/day
Hydrometallurgical processing capacity	500 kg/day
Surplus black mass available for sale	1,500 kg/day
Operating days	360 days/year
Capacity utilization in Year 1	70%
Full capacity utilization	From Year 2
Target project payback	During Year 4

The hydrometallurgical plant cost has been considered at **₹1.50 crore**, inclusive of process equipment, basic utilities, piping, electrical controls and installation associated with the 500 kg/day section.

2. Total Project CAPEX

A. Mechanical Recycling Plant

500 kg/hour Battery Recycling and Black Mass Production Line

Component	Description	Cost (₹ Cr)
Feeding system	Conveyor, feeding hopper and covered feeding belt	0.25
Primary shredding	Double-shaft shredder	0.80

Component	Description	Cost (₹ Cr)
Crushing system	Hammer mill and secondary crusher	0.40
Grinding system	Fine grinding and material liberation system	0.35
Separation system	Zigzag separator, air separator and vibrating screens	0.90
Magnetic separation	Drum and secondary magnetic separators	0.20
Dust collection system	Cyclones, pulse-jet bag filters and ducting	0.50
Electrical and control system	PLC panel, MCC, cabling and instrumentation	0.35
Structural installation	Platforms, supports, erection and commissioning	0.53
Total Mechanical Plant CAPEX		3.78

B. Hydrometallurgy Plant

500 kg/day Black Mass Processing

1. Leaching and Solid–Liquid Separation

Component	Cost (₹ Cr)
SS316L leaching reactors with agitators	0.22
Slurry preparation and leaching tanks	0.08
Chemical dosing and transfer pumps	0.05
Filter press and filtration arrangement	0.10
Settling and intermediate tanks	0.05
Subtotal	0.50

2. Purification and Solvent Extraction

Component	Cost (₹ Cr)
Purification and precipitation reactors	0.12
Compact mixer-settler system	0.20
Organic and aqueous storage tanks	0.08
Pumps, valves and process piping	0.10
Subtotal	0.50

3. Evaporation, Crystallization and Product Finishing

Component	Cost (₹ Cr)
Compact evaporator	0.12
Crystallizer	0.07
Centrifuge/solid separation	0.05
Product dryer	0.06
Product washing and packing system	0.05
Subtotal	0.35

4. Utilities, Electrical and Instrumentation

Component	Cost (₹ Cr)
Heating and steam arrangement	0.04
Cooling arrangement	0.03
Electrical panel and cabling	0.03
Instrumentation and automation	0.03
Installation and commissioning	0.02
Subtotal	0.15

Total Hydrometallurgy CAPEX

Section	Cost (₹ Cr)
Leaching and solid–liquid separation	0.50
Purification and solvent extraction	0.50
Evaporation and product finishing	0.35
Utilities, electrical and instrumentation	0.15
Total Hydrometallurgy Plant Cost	1.50

C. Land, Civil and Infrastructure

Component	Description	Cost (₹ Cr)
Industrial land	Approximately 2.5–3.0 acres	2.50
Land development	Levelling, boundary, internal preparation	0.75

Component	Description	Cost (₹ Cr)
Factory buildings and sheds	Mechanical, hydromet, storage and utility buildings	2.00
Internal roads and drainage	Roads, storm-water drainage and loading area	0.72
Total Land, Civil and Infrastructure Cost		5.97

D. Total Project Cost

Section	Cost (₹ Cr)
Mechanical recycling plant	3.78
Hydrometallurgy plant	1.50
Land, civil and infrastructure	5.97
Total Project CAPEX	11.25

Project Cost Summary

The total project cost is estimated at approximately **₹11.25 crore**, including land, buildings, mechanical recycling machinery, the 500 kg/day hydrometallurgical processing section and associated infrastructure.

The above cost excludes:

- Working capital margin;
- Financing charges during construction;
- GST input blockage, if any;
- Major statutory security deposits;
- Cost escalation beyond the implementation period.

3. Production and Material Allocation

Particular	Quantity
Battery scrap input	5,000 kg/day
Black mass recovery	2,000 kg/day
Black mass processed in hydrometallurgy	500 kg/day
Black mass available for direct sale	1,500 kg/day
Copper, aluminium and steel fractions	Approximately 1,250 kg/day

Particular	Quantity
Plastics, separator and other fractions	Approximately 1,000 kg/day
Moisture, electrolyte and process residue	Approximately 750 kg/day

4. Operating Expenditure

A. Raw Material Procurement Cost

Parameter	Value
Battery scrap purchase price	₹130/kg
Daily battery input	5,000 kg
Daily raw material cost	₹6.50 lakh/day
Annual raw material cost at 360 days ₹23.40 crore/year	

B. Mechanical Processing OPEX

Component	Cost (₹/day)
Electrical power	41,000
Operating manpower	5,000
Repairs and maintenance	12,000
Total Mechanical Processing OPEX	₹58,000/day

C. Hydrometallurgical Processing OPEX

The hydrometallurgical operating cost is approximately **₹245 per kg of black mass processed.**

1. Leaching

Component	Cost (₹/kg)
Sulphuric acid	35
Hydrogen peroxide/reducing agent	20
Process water and other reagents	16
Subtotal	71

2. Solvent Extraction and Purification

Component	Cost (₹/kg)
Solvent and extractant loss	25
Alkali and precipitation chemicals	20
Process operation and consumables	11
Subtotal	56

3. Power and Heating

Component	Cost (₹/kg)
Process electrical load	22
Auxiliary load	8
Heating	7
Subtotal	37

4. Evaporation and Product Finishing

Component	Cost (₹/kg)
Steam and thermal energy	40
Crystallization, drying and operation	19
Subtotal	59

5. Manpower

Component	Cost (₹/kg)
Hydromet operating manpower	21

Total Hydrometallurgical Cost

Parameter	Value
Total processing cost	₹244–245/kg
Daily hydromet feed	500 kg/day
Daily hydromet OPEX	₹1.225 lakh/day
Annual hydromet OPEX	₹4.41 crore/year

D. Total Daily OPEX

Component	Daily Cost
Raw material procurement	₹6.50 lakh
Mechanical processing	₹0.58 lakh
Hydrometallurgical processing	₹1.225 lakh
Total Daily OPEX	₹8.305 lakh/day

Rounded project operating cost:

₹8.31 lakh/day

5. Utility Requirements

A. Electrical Power

Section	Estimated Connected Load
Mechanical recycling line	300 kW
Hydrometallurgy section	50 kW
ETP, ECS and utilities	75 kW
Lighting and administration	20 kW
Total connected load	445 kW

Recommended Electrical Infrastructure

Parameter	Requirement
Recommended transformer	500–630 kVA
DG backup	250–320 kVA for essential loads
Power factor correction	Required
Separate emergency supply Safety, scrubber, fire and control systems	

B. Water Requirement

Application	Requirement
Hydromet process and chemical preparation	2.0 KLD
Cooling makeup	2.0 KLD

Application	Requirement
Equipment and floor washing	1.0 KLD
Scrubber makeup	0.5 KLD
Domestic use	1.5 KLD
Total Fresh Water Requirement	Approximately 7 KLD

A treated-water recovery system should be incorporated to reduce fresh-water demand.

C. Steam Requirement

Parameter	Requirement
Steam requirement	100–150 kg/hour
Application	Leaching, solution heating, evaporation and drying
Recommended steam generator	200 kg/hour

An electrical heating system or clean-fuel steam generator may be adopted depending on power availability and Pollution Control Board conditions.

D. Cooling Requirement

System	Capacity
Cooling tower	10–15 TR
Chiller, where required	5–7.5 TR

E. Compressed Air

Parameter	Requirement
Air compressor	10–15 HP
Uses	Pneumatic valves, bag-filter cleaning and instrumentation

F. Pollution-Control Utilities

Facility	Recommended Capacity
ETP	7.5 KLD

Facility	Recommended Capacity
RO/reuse system	5 KLD
Mechanical bag-filter airflow	Approximately 25,000 m ³ /hour
Hydromet wet scrubber	Approximately 10,000 m ³ /hour
Hazardous-waste storage	75–100 m ²

6. Revenue Assumptions

The project revenue is based on:

- Sale of black mass not processed internally;
- Sale of hydrometallurgically recovered products;
- Sale of copper, aluminium, steel and other scrap fractions.

No revenue from EPR certificate trading has been considered in the base financial model.

A. Hydrometallurgical Product Revenue

The earlier 2 TPD hydromet revenue assumption of ₹12.60 lakh/day has been proportionately adjusted for 500 kg/day processing.

Revenue Source	Daily Revenue
Hydrometallurgical recovered products	₹3.15 lakh/day

B. Black Mass Sale

Parameter	Value
Surplus black mass	1,500 kg/day
Assumed blended selling realization	Approximately ₹351/kg
Daily black mass revenue	₹5.26 lakh/day

The actual selling realization will depend on battery chemistry, cobalt and nickel content, moisture, impurities and prevailing metal prices.

C. Metal Scrap Revenue

Revenue Source	Daily Revenue
Copper, aluminium, steel and miscellaneous fractions	₹0.75 lakh/day

D. Total Revenue

Revenue Source	Daily Revenue
Hydromet products	₹3.15 lakh
Surplus black mass	₹5.26 lakh
Recovered metal scrap	₹0.75 lakh
Total Daily Revenue	₹9.16 lakh/day

Annual Revenue at Full Capacity

= ₹32.98 { crore/year}

7. Operating Profitability

Parameter	Value
Daily revenue	₹9.16 lakh
Daily OPEX	₹8.31 lakh
Daily operating surplus	₹0.85 lakh/day
Annual operating surplus	₹3.06 crore/year
Operating margin	Approximately 9.3%

The operating surplus represents earnings before depreciation, interest and tax.

8. Year-Wise Financial Projection

Capacity Utilization

Year	Capacity Utilization
Year 1	70%
Year 2	100%
Year 3	100%
Year 4	100%
Year 5	100%

Revenue, OPEX and Operating Surplus

Year	Capacity	Revenue (₹ Cr)	OPEX (₹ Cr)	Operating Surplus (₹ Cr)
Year 1	70%	23.08	20.93	2.14
Year 2	100%	32.98	29.90	3.06
Year 3	100%	32.98	29.90	3.06
Year 4	100%	32.98	29.90	3.06
Year 5	100%	32.98	29.90	3.06

9. Project Payback and ROI

Year	Operating Cash Surplus (₹ Cr)	Cumulative Position (₹ Cr)
Initial project investment	—	-11.25
Year 1	2.14	-9.11
Year 2	3.06	-6.05
Year 3	3.06	-2.99
Year 4	3.06	+0.07
Year 5	3.06	+3.13

Payback Period

The project crosses cumulative positive operating cash flow during:

Year 4

Indicative payback period:

Approximately 4 years

10. Financial Indicators

Indicator	Estimated Value
Total project investment	₹11.25 crore
Annual revenue at full capacity	₹32.98 crore
Annual OPEX at full capacity	₹29.90 crore
Annual operating surplus	₹3.06 crore
Operating margin	Approximately 9.3%
Simple annual return on project cost	Approximately 27.2%
Payback achievement	Year 4
Full capacity achievement	Year 2

The simple annual return stated above is based on steady-state operating surplus before depreciation, interest and tax. The post-tax return will depend on the final debt-equity structure, applicable interest rate, depreciation schedule and taxation.

11. Sensitivity Analysis

A. Black Mass Selling Price

Scenario	Indicative Impact
Selling price decreases by 10%	Project payback may extend beyond Year 5
Selling price decreases by 5%	Project payback may move toward late Year 4 or Year 5
Selling price increases by 5%	Payback may improve to approximately 3.5 years

B. Raw Material Cost

Scenario	Indicative Impact
Battery scrap cost increases by ₹5/kg	Annual cost increases by approximately ₹0.90 crore
Battery scrap cost increases by ₹10/kg	Annual cost increases by approximately ₹1.80 crore

Scenario	Indicative Impact
Battery scrap cost decreases by ₹5/kg	Annual operating surplus improves by approximately ₹0.90 crore

C. Hydromet Recovery

Scenario	Impact
Improved recovery and purity	Increases product realization and profitability
Lower cobalt/nickel content	Reduces hydromet revenue
High LFP proportion in feed	Reduces black mass value and metal recovery revenue

12. Key Financial Conditions

Achievement of the projected Year 4 payback depends on:

1. Maintaining the battery scrap procurement price near ₹130/kg;
2. Achieving approximately 40% black mass recovery;
3. Maintaining average surplus black mass realization near ₹350–351/kg;
4. Achieving the assumed hydrometallurgical metal recovery efficiency;
5. Operating at 70% capacity in Year 1 and 100% from Year 2;
6. Maintaining 360 operating days annually;
7. Avoiding prolonged plant shutdowns;
8. Maintaining segregation of high-value NMC/LCO and lower-value LFP battery feed;

13. Conclusion

The proposed facility, comprising a **500 kg/hour mechanical lithium-ion battery recycling line and a 500 kg/day hydrometallurgical processing section**, requires an estimated total project investment of approximately **₹11.25 crore**, including land and civil infrastructure.

At full operational capacity, the project is expected to:

- Process 5 TPD of lithium-ion battery waste;
- Produce approximately 2 TPD of black mass;
- Process 0.5 TPD black mass through hydrometallurgy;
- Sell approximately 1.5 TPD surplus black mass;
- Generate annual revenue of approximately ₹32.98 crore;
- Incur annual operating expenditure of approximately ₹29.90 crore;
- Generate an annual operating surplus of approximately ₹3.06 crore.

Based on 70% utilization in Year 1 and full utilization from Year 2, cumulative operating cash flow becomes positive during the **fourth year of operation**. The project therefore demonstrates technical feasibility and acceptable financial viability, subject to feed chemistry, raw material price, product realization and operating-efficiency assumptions.

17. Financial Projections

FIVE-YEAR FINANCIAL PROJECTIONS

500 kg/hour Mechanical Recycling Plant with 500 kg/day Hydrometallurgy Plant

1. Key Financial Assumptions

Parameter	Assumption
Total project CAPEX	₹11.25 crore
Battery recycling capacity	5,000 kg/day
Black mass generation	2,000 kg/day
Hydrometallurgy feed	500 kg/day
Surplus black mass sale	1,500 kg/day
Operating days	360 days/year
Capacity utilization in Year 1	70%
Capacity utilization from Year 2	100%
Battery scrap purchase price	₹130/kg
Full-capacity annual revenue	₹32.98 crore
Full-capacity annual OPEX	₹29.90 crore
Full-capacity EBITDA	₹3.08 crore
Depreciable project cost	₹8.75 crore
Annual depreciation	₹0.875 crore
Income-tax rate	25%
EPR certificate revenue	Not considered
Subsidy income	Not considered
Financing cost	Not considered; depends on debt structure

The financial projections are presented on an unlevered project basis. Interest expense and loan repayment may be incorporated after finalization of the debt-equity structure and lending terms.

2. Capacity Utilization

Financial Year Capacity Utilization

Year 1	70%
Year 2	100%
Year 3	100%
Year 4	100%
Year 5	100%

The plant is expected to stabilize during the first year and achieve full commercial capacity from the second year onward.

3. Revenue Projection

Full-Capacity Daily Revenue

Revenue Source	Daily Revenue
Hydrometallurgical recovered products	₹3.15 lakh
Sale of 1,500 kg/day surplus black mass	₹5.26 lakh
Sale of copper, aluminium, steel and other fractions	₹0.75 lakh
Total Daily Revenue	₹9.16 lakh

Year-Wise Revenue

Revenue Source	Year 1	Year 2	Year 3	Year 4	Year 5
Hydromet products	7.94	11.34	11.34	11.34	11.34
Surplus black mass	13.26	18.94	18.94	18.94	18.94
Recovered metal fractions	1.89	2.70	2.70	2.70	2.70
Total Revenue	23.09	32.98	32.98	32.98	32.98

Figures are in ₹ crore.

4. Operating Expenditure Projection

Full-Capacity Annual OPEX

Cost Component	Annual Cost
Battery scrap procurement	₹23.40 crore
Mechanical processing	₹2.09 crore
Hydrometallurgical processing	₹4.41 crore
Total Annual OPEX	₹29.90 crore

Year-Wise OPEX

Cost Component	Year 1	Year 2	Year 3	Year 4	Year 5
Raw material procurement	16.38	23.40	23.40	23.40	23.40
Mechanical processing	1.46	2.09	2.09	2.09	2.09
Hydromet processing	3.09	4.41	4.41	4.41	4.41
Total OPEX	20.93	29.90	29.90	29.90	29.90

Figures are in ₹ crore.

5. Projected Profit and Loss Statement

Particular	Year 1	Year 2	Year 3	Year 4	Year 5
Total revenue	23.09	32.98	32.98	32.98	32.98
Operating expenditure	20.93	29.90	29.90	29.90	29.90
EBITDA	2.16	3.08	3.08	3.08	3.08
Depreciation	0.88	0.88	0.88	0.88	0.88
EBIT	1.28	2.20	2.20	2.20	2.20
Interest expense	—	—	—	—	—
Profit Before Tax	1.28	2.20	2.20	2.20	2.20
Income tax at 25%	0.32	0.55	0.55	0.55	0.55
Profit After Tax	0.96	1.65	1.65	1.65	1.65

Figures are in ₹ crore and may vary slightly due to rounding.

6. Cash Accrual Projection

Cash accrual represents profit after tax plus non-cash depreciation.

Particular	Year 1	Year 2	Year 3	Year 4	Year 5
Profit after tax	0.96	1.65	1.65	1.65	1.65
Add: Depreciation	0.88	0.88	0.88	0.88	0.88
Annual Cash Accrual	1.84	2.53	2.53	2.53	2.53
Cumulative Cash Accrual	1.84	4.37	6.90	9.43	11.96

Figures are in ₹ crore.

The post-tax cash accrual over five years is approximately **₹11.96 crore**, before considering financing costs and loan repayments.

7. Operating Payback Analysis

For project-level payback, EBITDA has been considered as the operating cash surplus before depreciation, financing and taxation.

Year	Annual Operating Surplus	Cumulative Operating Surplus	Balance Project Cost
Initial investment	—	—	₹11.25 Cr
Year 1	2.16	2.16	9.09
Year 2	3.08	5.24	6.01
Year 3	3.08	8.32	2.93
Year 4	3.08	11.40	Surplus ₹0.15 Cr
Year 5	3.08	14.48	Surplus ₹3.23 Cr

Payback Period

The cumulative operating surplus exceeds the initial project investment during the fourth year.

Indicative operating payback: approximately 3.9 to 4 years

This payback represents recovery of project investment from operating cash generation before interest, depreciation and taxation.

8. Profitability Indicators

Indicator	Projected Value
Full-capacity annual revenue	₹32.98 crore
Full-capacity annual EBITDA	₹3.08 crore
EBITDA margin	9.34%
Full-capacity PAT	₹1.65 crore
Net profit margin	Approximately 5.0%
Annual post-tax cash accrual	₹2.53 crore
Simple EBITDA return on project cost	Approximately 27.4%
Simple PAT return on project cost	Approximately 14.7%
Operating payback	During Year 4

9. Five-Year Financial Summary

Particular	Year 1	Year 2	Year 3	Year 4	Year 5
Capacity utilization	70%	100%	100%	100%	100%
Revenue	23.09	32.98	32.98	32.98	32.98
OPEX	20.93	29.90	29.90	29.90	29.90
EBITDA	2.16	3.08	3.08	3.08	3.08
EBITDA margin	9.35%	9.34%	9.34%	9.34%	9.34%
Profit before tax	1.28	2.20	2.20	2.20	2.20
Profit after tax	0.96	1.65	1.65	1.65	1.65
Cash accrual	1.84	2.53	2.53	2.53	2.53

Financial figures are in ₹ crore.

10. Break-Even Analysis

Fixed Costs

Fixed-Cost Component	Estimated Annual Cost
Depreciation	₹0.88 crore
Fixed salaries and administration included in OPEX	Approximately ₹0.60–0.75 crore
Insurance, compliance and fixed maintenance	Approximately ₹0.20–0.30 crore
Estimated Fixed Cost	Approximately ₹1.70–1.90 crore/year

Based on the projected contribution margin, the operating break-even level is estimated at approximately:

60%–65% capacity utilization

The plant is projected to operate above the break-even level during Year 1 itself, based on 70% capacity utilization.

11. Sensitivity Analysis

Scenario	Financial Impact
Battery scrap cost increases by ₹5/kg	Annual EBITDA reduces by approximately ₹0.90 crore
Battery scrap cost increases by ₹10/kg	Annual EBITDA reduces by approximately ₹1.80 crore
Black mass selling price falls by 5%	Payback may shift toward late Year 4 or Year 5
Black mass selling price falls by 10%	Project profitability may become marginal
Product realization increases by 5%	Annual revenue improves by approximately ₹1.65 crore
Hydromet recovery improves	Higher saleable metal output and improved EBITDA
NMC/LCO proportion increases	Improved black mass and hydromet product value
LFP proportion increases	Lower average black mass realization

12. Key Financial Risks

The projected financial performance remains sensitive to:

- Battery chemistry and metal content;

- Raw material procurement price;
- Black mass selling realization;
- Cobalt, nickel and lithium market prices;
- Hydrometallurgical recovery efficiency;
- Chemical consumption;
- Plant capacity utilization;
- Electricity and fuel tariffs;
- Disposal cost of hazardous residues;
- Product purity and buyer acceptance.

Long-term raw material supply agreements and product-offtake arrangements should be executed before commercial commissioning.

13. Financial Conclusion

The proposed **500 kg/hour lithium-ion battery mechanical recycling plant with a 500 kg/day hydrometallurgical section** is projected to generate annual revenue of approximately **₹32.98 crore** at full capacity.

The project is expected to generate:

- EBITDA of approximately **₹3.08 crore per year**;
- Profit after tax of approximately **₹1.65 crore per year**;
- Annual post-tax cash accrual of approximately **₹2.53 crore** at stabilized operations;
- Cumulative operating surplus exceeding the initial investment during the **fourth year**.

The projections do not include revenue from EPR certificates, government subsidies, grants or carbon credits. Any benefit realized from these sources will improve the project's overall financial returns.

18. Manpower Requirements

1. Overall Manpower Summary

Category	No. of Personnel
Management & Admin	5–6
Mechanical Plant	8–10
Hydromet Plant	12–15
Utilities & ETP	5–6
Maintenance & Lab	6–8
Total Manpower	36–45 persons

2. Mechanical Section (Black Mass Plant)

Role	Nos.	Responsibility
Supervisor	1	Overall operation
Machine operators	3–4	Shredder, crusher, separator
Helpers / loaders	3–4	Feeding, material handling
Electrician (shared)	1	Equipment support
Total	8–10	

Mostly **semi-skilled workforce**

3. Hydrometallurgy Section (Core Team)

A. Process Operation

Role	Nos.
Plant manager / metallurgist	1
Shift engineers	2
Reactor operators	4
SX operators	2
EW operators	2
Helpers	4
Subtotal	15

B. Laboratory & Quality Control

Role	Nos.
Chemist	2
Lab technicians	2
Subtotal	4

4. Utilities & ETP Section

Role	Nos.	Responsibility
Utility operator	2	Boiler, cooling tower
ETP operator	2	Effluent treatment
Helper	1–2	Support
Total	5–6	

5. Maintenance Team

Role	Nos.
Mechanical engineer	1
Fitters	2
Electrician	2
Instrument technician	1
Total	6

6. Admin & Support Staff

Role	Nos.
Plant head / GM	1
Accounts	1
Procurement	1
HR / Admin	1
Security	2
Total	6

7. Shift Planning

Recommended:

- **Single shift (mechanical)**
- **2 shifts (hydromet)**

Section	Shift
Mechanical	1 shift
Hydromet	2 shifts
Utilities	Continuous

8. Manpower Cost Estimate

Category	Monthly Cost
Skilled staff	₹6–8 lakh
Semi-skilled	₹3–4 lakh
Admin & support	₹2–3 lakh
Total Monthly Cost	₹10–12 lakh/month

Annual manpower cost:

₹1.2 – 1.5 Cr/year

9. Manpower Cost Contribution

Parameter	Value
% of total OPEX	~10%
Cost per kg (approx.)	₹20–25/kg (hydromet included)

The plant will employ approximately 36–45 personnel across mechanical processing, hydrometallurgical operations, utilities, maintenance, and administration. The manpower structure includes skilled technical staff such as metallurgists and chemists, along with semi-skilled operators and support staff. The estimated annual manpower cost is approximately ₹1.2–1.5 crore, contributing around 10% of the total operating cost.

19. Hazardous waste handling

1. Types of Hazardous Waste Generated

A. Mechanical Section

Waste Type	Source	Nature
Electrolyte residues	Battery shredding	Flammable, toxic
Fine dust (black mass loss)	Crushing/separation	Heavy metal containing
Plastic separators	Battery dismantling	Contaminated waste

B. Hydromet Section

Waste Type	Source	Nature
Sludge (metal hydroxides)	ETP	Hazardous (Co, Ni, Mn)
Spent chemicals	Leaching/SX	Acidic
Solvent residues	SX process	Organic hazardous
Filter cake	Filtration	Heavy metal bearing

C. Utilities & ETP

Waste Type	Source
ETP sludge	Effluent treatment
Evaporation salts	ZLD system
Oil & grease waste	Machinery

2. Quantity Estimation (Indicative)

Waste Type	Quantity
ETP sludge	50–100 kg/day
Filter cake	100–200 kg/day
Evaporation salts	50–100 kg/day
Dust waste	20–50 kg/day

3. Waste Handling Strategy

A. Segregation at Source

- Separate:
 - Solid waste
 - Liquid waste
 - Chemical waste
- Dedicated labeled containers

B. Storage System (As per CPCB norms)

Design Requirements:

- Impermeable flooring
- Covered shed
- Leachate collection system
- Fire-resistant construction

Storage Categories:

Waste	Storage Method
Sludge	HDPE-lined containers
Chemical waste	FRP/HDPE tanks
Solvent waste	Sealed drums
Dust	Bagged storage

C. Treatment Methods

1. Effluent Treatment (ETP)

- Neutralization
- Precipitation of metals
- Sludge separation

Output:

- Treated water (reused)
- Sludge (hazardous)

2. Zero Liquid Discharge (ZLD)

- Multi-effect evaporator
- No liquid discharge outside plant

3. Solvent Recovery

- Reuse of SX solvents
- Minimizes waste generation

4. Dust Recovery

- Collected via bag filters
- Recycled back into process (where possible)

4. Disposal Method

Authorized Disposal Only

Waste	Disposal Method
ETP sludge	TSDf (secured landfill)
Chemical residues	Authorized hazardous waste recycler
Oil waste	Registered recycler
Plastic waste	Co-processing (cement plants)

Must comply with:

- **Hazardous Waste Management Rules, 2016**

5. Regulatory Compliance

Required:

- Authorization from **SPCB**
- Hazardous waste registration
- Annual return filing
- Waste manifest system

Documentation:

- Waste generation log
- Disposal records
- Transport manifest (Form 10)

6. Safety Measures

Fire & Chemical Safety

- Fire-resistant storage
- Spill containment systems
- Emergency showers

PPE Requirements

- Chemical-resistant gloves
- Respirators
- Safety suits

Monitoring

- Periodic soil/water testing
- Stack emission monitoring

7. On-Site Layout Provision

- Dedicated **hazardous waste storage zone**
- Located:
 - Away from production
 - Near ETP

20. Regulatory compliance Requirement

1. Key Applicable Laws & Regulations

A. Battery Waste Management Rules, 2022

(Most important regulation)

Requirements:

- Mandatory registration on **CPCB portal**
- Operate only as **authorized recycler**
- Maintain traceability of batteries
- Issue recycling certificates (for producers)

B. Hazardous Waste Management Rules, 2016

Applicable because:

- Black mass, sludge, chemicals = hazardous waste

Requirements:

- Authorization from **SPCB**
- Proper storage, handling & disposal
- Tie-up with **TSDF facility**

C. Environment Protection Act, 1986

- Umbrella law covering pollution control
- Compliance through:
 - Air Act
 - Water Act

2. Pollution Control Approvals (Mandatory)

A. Consent to Establish (CTE)

- Required **before construction**
- Issued by: **State Pollution Control Board (SPCB)**

B. Consent to Operate (CTO)

- Required **before plant operation**
- Valid for 3–5 years

C. Air & Water Consent

Act	Requirement
Air Act, 1981	Emission control (ECS)
Water Act, 1974	Effluent treatment (ETP/ZLD)

4. CPCB Registration (Battery Recycling)

Mandatory for:

- All recyclers

Requirements:

- Plant details
- Capacity
- Process technology
- Pollution control systems

5. Factory & Industrial Licenses

A. Factory License

- Under **Factories Act, 1948**
- Issued by State Labour Department

B. Trade License

- From local authority / municipality

C. MSME Registration (Optional but recommended)

- For subsidy and financing benefits

6. Utility & Infrastructure Approvals

Approval	Authority
Electricity connection	DISCOM
Boiler license	Boiler Inspectorate
Fire NOC	Fire Department
Water extraction	Groundwater Authority (if borewell)

7. Safety & Hazard Compliance

A. Fire Safety

- Fire NOC required
- Special provisions for:
 - Battery storage
 - Chemical handling

B. Chemical Storage Compliance

- Follow **MSDS guidelines**
- Proper labeling & storage

C. Occupational Safety

- PPE compliance
- Safety training
- Emergency response plan

8. Waste Management Compliance

A. Hazardous Waste Authorization

- For:
 - Sludge
 - Chemical waste

B. TSDF Tie-up

- Mandatory for disposal

C. Manifest System

- Track waste movement

9. ESG & Sustainability (Optional but Strong)

- ISO 14001 (Environmental)
- ISO 45001 (Safety)
- Carbon footprint reporting

10. Compliance Timeline (Important for DPR)

Stage	Approval
Pre-construction	CTE, Land approval
Construction	Factory registration
Pre-operation	CTO, CPCB registration
Operation	Annual returns, audits

The proposed project will comply with all applicable environmental and industrial regulations, including the Battery Waste Management Rules, 2022, Hazardous Waste Management Rules, 2016, and the Environment Protection Act, 1986. Necessary approvals such as Consent to Establish, Consent to Operate, and CPCB registration will be obtained prior to commencement of operations. The plant will incorporate advanced pollution control systems and adhere to strict safety and waste management protocols to ensure full regulatory compliance.

21. POTENTIAL VALUE-ADDED PRODUCTS FROM BY-PRODUCTS

5 TPD Lithium-Ion Battery Recycling Plant with 500 kg/day Hydrometallurgy

The proposed plant will generate black mass, non-ferrous metal fractions, ferrous scrap, plastics, separator material, graphite-rich residue, chemical salts and process residues. Revenue can be increased by upgrading suitable by-products into standardized industrial products instead of selling all recovered fractions as mixed scrap.

Battery recycling is increasingly being treated as a source of secondary copper, lithium, nickel, cobalt and graphite. However, the feasibility of each value-added product depends on feed chemistry, purity, contamination level, additional CAPEX and market qualification. (IEA)

1. By-Product Availability

Based on the project mass balance:

By-product or Intermediate Stream	Approximate Daily Quantity
Surplus black mass	1,500 kg/day
Copper, aluminium and steel fractions	1,250 kg/day combined
Plastics and separator fractions	1,000 kg/day
Recovered graphite from hydrometallurgy	Approximately 142.5 kg/day
Hydromet residue and sludge	Approximately 102 kg/day
Electrolyte, moisture and contaminated fractions	Part of approximately 750 kg/day process stream

The metal fractions should be further segregated into copper, aluminium and steel before revenue calculations are finalized.

2. Surplus Black Mass Upgradation

2.1 Standardized Black Mass Concentrate

Instead of selling mixed black mass directly after mechanical separation, it can be upgraded through:

- Fine screening;
- Removal of copper and aluminium contamination;
- Magnetic separation;

- Controlled drying;
- Moisture reduction;
- Homogenization;
- Batch-wise chemical assay;
- Sealed packaging.

Saleable Product

Assay-certified low-moisture black mass concentrate

Commercial Benefit

A product with controlled moisture, low metallic contamination and verified cobalt, nickel, lithium and graphite content generally commands a stronger realization than ungraded mixed powder.

Required Additional Equipment

Equipment	Purpose
Fine vibrating screen	Particle-size control
Secondary magnetic separator	Residual iron removal
Eddy-current/air separator	Aluminium and copper removal
Low-temperature dryer	Moisture control
Blender/homogenizer	Uniform batch composition
Laboratory analyser	Metal assay and quality certificate
Automatic packing system	Sealed drum or bag packing

Priority

Very high

This is the simplest value-addition opportunity because the plant is already producing black mass.

2.2 Additional Hydrometallurgical Processing

The plant currently processes only 500 kg/day of the approximately 2,000 kg/day black mass generated. Future expansion of the hydromet section can convert the remaining black mass into higher-value products.

Potential products include:

- Cobalt sulphate;

- Nickel sulphate;
- Manganese sulphate;
- Lithium carbonate;
- Mixed nickel-cobalt-manganese hydroxide;
- Graphite concentrate.

This provides substantially higher value addition but requires higher chemical consumption, stronger ETP/ZLD infrastructure and product-quality control.

Priority

High as a future expansion, subject to secured feed chemistry and product buyers.

3. Copper-Based Products

Copper is recovered mainly from current collectors, busbars, cables and terminals.

3.1 Clean Copper Granules

Mixed copper foil and wiring can be processed into:

Clean copper granules or chopped copper

Process

Copper-rich fraction



Granulation



Air separation



Removal of plastic and aluminium



Clean copper granules

Potential Buyers

- Copper smelters;
- Cable manufacturers;
- Brass manufacturers;
- Foundries;
- Metal traders.

Required Equipment

- Cable granulator;
- Fine crusher;
- Air separator;
- Vibrating screen;
- Dust collector.

Priority

Very high

This is generally more practical than manufacturing finished copper wire or cathode copper within the same facility.

3.2 Copper Briquettes

Fine copper foil and copper powder can be compacted into dense briquettes.

Advantages

- Easier storage;
- Reduced transportation volume;
- Lower handling losses;
- Better furnace charging;
- Higher acceptance by smelters and foundries.

Required Equipment

- Hydraulic metal briquetting press;
- Drying and screening system.

Priority

High

3.3 Copper Oxide or Copper Salts

Fine copper-bearing material can technically be processed into:

- Copper oxide;
- Copper sulphate;
- Copper hydroxide.

However, this requires chemical processing, purification and additional wastewater treatment. It should be considered only where a consistent copper-bearing stream and confirmed buyer are available.

Priority

Medium to low

4. Aluminium-Based Products

Aluminium is recovered from current collectors, casings, cooling plates and structural battery-pack components.

4.1 Clean Aluminium Scrap or Granules

Separated aluminium can be cleaned, screened and sold as:

- Aluminium foil scrap;
- Aluminium granules;
- Aluminium twitch-type scrap;
- Segregated casting-grade scrap.

Copper and aluminium foil separation is commercially important because mixed fine foils generally attract lower realization. ([ScienceDirect](#))

Priority

Very high

4.2 Aluminium Briquettes

Fine aluminium foil and powder can be compressed into briquettes for sale to secondary aluminium smelters.

Benefits

- Improved bulk density;
- Lower oxidation during handling;
- Reduced dust;
- Lower freight cost;
- Better furnace charging.

Safety Requirement

Fine aluminium dust is combustible. Briquetting must include:

- Dust extraction;
- Earthing;
- Spark control;
- Moisture control;
- Explosion-risk assessment.

Priority

High

4.3 Aluminium Ingots

The plant may install a secondary aluminium melting furnace to manufacture ingots.

However, this is not recommended in the initial project because it would require:

- Additional furnace CAPEX;
- Separate air pollution control;
- Fuel consumption;
- Dross management;
- Metallurgical quality control;
- Additional consent conditions.

Priority

Future option only

5. Steel and Ferrous Products

5.1 Baled Ferrous Scrap

Steel casings, frames and structural fractions may be processed into:

Baled steel scrap

Required Equipment

- Magnetic separator;
- Hydraulic baling press;
- Weighing and bundling system.

Benefits

- Reduced storage space;
- Lower transport cost;
- Improved saleability to steel melting units.

Priority

High

5.2 Shredded Steel Scrap

Where the steel stream is sufficiently clean, it can be sized and sold as furnace-ready shredded scrap.

Priority

Medium

6. Plastic and Separator Products

The plastic stream may contain:

- PP;
- PE;
- ABS;
- Nylon;
- PVC traces;
- Battery casings;
- Separator films;
- Electrolyte contamination;
- Adhesives and binders.

The stream must first be segregated and tested because contaminated separators and mixed battery plastics may not be suitable for ordinary plastic recycling.

6.1 Recycled Plastic Granules

Clean and chemistry-segregated plastics may be converted into:

- PP granules;
- PE granules;
- ABS regrind;
- Mixed engineering-plastic compounds.

Process

Plastic fraction



Manual/optical sorting



Washing



Drying



Grinding



Extrusion and pelletization

↓
Recycled plastic granules
End Uses

- Non-critical moulded components;
- Pallets;
- Crates;
- Industrial bins;
- Cable ducts;
- Plastic boards;
- Utility products.

Important Condition

Only clean and non-hazardous plastic fractions should be processed. Electrolyte-contaminated separators should not be mixed into ordinary recycled-plastic products without characterization and regulatory acceptance.

Priority

High for clean casing plastics; low for contaminated separators

6.2 Plastic Boards, Pallets and Utility Products

Lower-grade mixed plastic granules can be converted into:

- Plastic pallets;
- Industrial boards;
- Dunnage blocks;
- Cable covers;
- Non-structural profiles;
- Waste-storage bins.

This option may provide better value than selling mixed plastic scrap, particularly where color and cosmetic appearance are not important.

Priority

Medium to high

6.3 Refuse-Derived Fuel or Co-processing Feed

Non-recyclable plastic fractions may be evaluated for authorized co-processing in cement kilns.

They should not automatically be treated as fuel because the material may contain:

- Fluorinated binder;
- Electrolyte contamination;
- Heavy-metal traces;
- Halogenated plastics.

Use as co-processing feed requires characterization and acceptance by an authorized facility.

Priority

Disposal/recovery route, not a principal product

7. Graphite-Based Products

Graphite recovery is one of the most promising future value-addition opportunities. Recycling innovation is increasingly focusing not only on lithium, nickel and cobalt but also on graphite and copper recovery.

7.1 Industrial-Grade Recovered Graphite

The graphite-rich filter cake can be processed through:

- Washing;
- Neutralization;
- Filtration;
- Drying;
- Milling;
- Screening;
- Magnetic impurity removal.

Potential Product

Recovered graphite concentrate

Potential Applications

- Foundry carbon additive;
- Refractory applications;
- Friction materials;
- Conductive fillers;
- Lubricant and industrial carbon applications.

Priority

High

7.2 Conductive Carbon Additive

After purification and particle-size control, recovered graphite may be blended or processed for use as:

- Conductive additive in polymers;
- Conductive coatings;
- Carbon masterbatch;
- Antistatic compounds.

Additional Requirements

- Controlled particle size;
- Low metal impurity;
- Low moisture;
- Consistent electrical conductivity;
- Customer qualification.

Priority

Medium

7.3 Battery-Grade Regenerated Graphite

Recovered graphite can potentially be purified and regenerated for battery-anode applications. Research has shown that purified spent graphite can retain useful electrochemical performance, but industrial commercialization requires stringent purification and qualification. ([ScienceDirect](#))

Required Processing

- Acid purification;
- High-temperature treatment;
- Carbon coating;
- Spheroidization;
- Classification;
- Surface modification;
- Electrochemical testing.

Commercial Position

This is a high-value product but requires significant additional CAPEX, R&D and qualification by cell manufacturers.

Priority

Long-term strategic opportunity

8. Hydrometallurgical Product Upgradation

8.1 Cobalt Sulphate

Cobalt may be sold as:

- Cobalt sulphate solution;
- Cobalt sulphate crystals;
- Cobalt hydroxide intermediate;
- Cobalt metal equivalent.

Crystallized cobalt sulphate generally offers better marketability than low-grade mixed precipitate, provided purity is maintained.

8.2 Nickel Sulphate

Nickel may be recovered as:

- Nickel sulphate solution;
- Nickel sulphate crystals;
- Nickel hydroxide;
- Nickel metal through electrowinning.

Battery-grade material requires strict impurity limits and buyer approval.

8.3 Manganese Sulphate Monohydrate

Manganese may be crystallized into:

Manganese sulphate monohydrate

Potential buyers include:

- Battery-material manufacturers;
- Chemical manufacturers;
- Micronutrient manufacturers, subject to applicable specifications.

Battery-derived material should not be sold into agriculture or feed applications without full regulatory and quality compliance.

8.4 Lithium Carbonate

Lithium-bearing solution may be converted into:

- Technical-grade lithium carbonate;
- Battery-grade lithium carbonate after advanced purification.

Value-Addition Steps

- Removal of sodium, calcium and magnesium;
- Concentration;
- Carbonate precipitation;
- Washing;
- Drying;
- Micronization;
- Quality certification.

Priority

Very high

8.5 Mixed NCM Hydroxide or Carbonate

Instead of separately recovering cobalt, nickel and manganese, the plant may produce a mixed intermediate such as:

- Mixed nickel-cobalt-manganese hydroxide;
- Mixed NCM carbonate.

Advantages

- Fewer separation stages;
- Lower solvent-extraction complexity;
- Potentially lower OPEX;
- Suitable as feed for downstream refiners.

Limitation

The required nickel-cobalt-manganese ratio varies by buyer and battery chemistry.

Priority

High where an offtake buyer is available

9. Sodium Sulphate and Other Process Salts

Hydrometallurgical operations may generate sodium sulphate or similar salts from:

- Neutralization;
- Alkali dosing;
- Lithium precipitation;
- Evaporation of process liquor.

Potential Product

Technical-grade sodium sulphate

Possible Uses

- Detergent manufacturing;
- Glass manufacturing;
- Textile processing;
- Pulp and paper.

Important Condition

The salt must meet impurity limits. Heavy-metal-contaminated evaporator salts cannot be sold as commercial products and must be disposed of through an authorized route.

Priority

Medium, subject to laboratory quality

10. Metal Recovery from ETP Sludge

ETP sludge may contain residual:

- Cobalt;
- Nickel;
- Manganese;
- Copper;
- Lithium.

Instead of directly sending all sludge to TSDF, high-metal-bearing sludge may be:

- Returned to the leaching circuit;
- Reprocessed in a dedicated recovery batch;
- Sent to an authorized metal-recovery facility.

Commercial Benefit

- Reduced metal loss;
- Reduced hazardous-waste disposal cost;
- Improved overall recovery efficiency.

Restriction

ETP sludge should not be marketed as an ordinary product. Recovery or transfer must be undertaken only within the permitted hazardous-waste and battery-recycling framework.

Priority

High as internal process recovery

11. Products Not Recommended Without Further Study

The following products should not be included as assured revenue streams at the initial stage:

Proposed Product	Reason
Fertilizer from hydromet salts	Heavy-metal contamination and regulatory risk
Fuel from contaminated separator plastic	Fluoride, electrolyte and emission concerns
Direct reuse of untreated graphite in new batteries	Purity and electrochemical qualification requirements
Sale of ETP sludge	Hazardous-waste restrictions
Sale of mixed plastic without characterization	Possible electrolyte and metal contamination
Recovery of electrolyte solvent for resale	Complex purification, safety and quality requirements
Aluminium ingots	Additional furnace, APCS and dross-management requirements

12. Recommended Revenue-Enhancement Plan

Phase 1: Immediate Implementation

Product	Additional CAPEX Level	Commercial Priority
Assay-certified black mass	Low	Very high

Product	Additional CAPEX Level Commercial Priority
Clean copper granules	Low–medium Very high
Copper briquettes	Low High
Clean aluminium granules	Low–medium Very high
Aluminium briquettes	Low High
Baled steel scrap	Low High
Industrial-grade recovered graphite	Medium High

Phase 2: After Plant Stabilization

Product	Additional CAPEX Level Commercial Priority
Recycled PP/PE/ABS granules	Medium High
Plastic pallets and boards	Medium Medium
Crystallized nickel sulphate	Medium High
Crystallized cobalt sulphate	Medium High
Technical-grade lithium carbonate	Medium Very high
Manganese sulphate	Medium High
Sodium sulphate recovery	Medium Conditional

Phase 3: Strategic Expansion

Product	Additional CAPEX Level Commercial Priority
Battery-grade lithium carbonate	High Very high
Battery-grade nickel/cobalt sulphates	High Very high
Mixed NCM precursor intermediate	High High
Regenerated battery-grade graphite	High Strategic
Full hydromet processing of 2 TPD black mass	High Very high

13. Recommended Product Mix

The most practical product mix for improving revenue without creating excessive technical complexity is:

1. **Assay-certified black mass concentrate** from the 1.5 TPD surplus stream;
2. **Clean copper granules or briquettes;**
3. **Clean aluminium granules or briquettes;**
4. **Baled steel scrap;**
5. **Recovered industrial-grade graphite;**
6. **Technical-grade lithium carbonate;**
7. **Nickel, cobalt and manganese sulphates or mixed NCM intermediate;**
8. **Recycled plastic granules only from clean casing plastics.**

14. Revenue-Enhancement Conclusion

The highest immediate revenue improvement can be achieved by upgrading the existing mechanical outputs rather than adding unrelated downstream processes. Priority should therefore be given to:

- Improving black mass purity and assay certification;
- Separating copper and aluminium more efficiently;
- Converting loose metal foils into granules or briquettes;
- Recovering saleable graphite concentrate;
- Crystallizing hydromet products into consistent commercial grades;
- Recovering valuable metals from internal process residues;
- Segregating clean plastic casing material for granule production.

The future expansion of the hydrometallurgical section from **500 kg/day toward the full 2,000 kg/day black mass output** offers the largest long-term value-addition opportunity. However, this expansion should be undertaken only after securing feed-quality control, chemical-product offtake agreements, adequate ETP/ZLD capacity and product-purity guarantees.

22. Conclusion

The proposed **5 TPD Lithium-Ion Battery Recycling Plant integrated with a 500 kg/day Hydrometallurgical Metal Recovery Unit** represents a technically feasible, environmentally sustainable, and commercially viable investment aligned with India's rapidly expanding electric vehicle ecosystem and circular economy objectives. With the implementation of the Battery Waste Management Rules, 2022, increasing localization of battery manufacturing, and growing demand for critical minerals such as lithium, cobalt, nickel, and manganese, the project is strategically positioned to cater to the emerging domestic recycling industry while reducing dependence on imported raw materials.

The proposed facility has been designed to process **5 metric tonnes of waste lithium-ion batteries per day**, generating approximately **2 TPD of black mass**, of which **500 kg/day** will be processed through the hydrometallurgical section for recovery of high-value metals, while the remaining **1.5 TPD** of black mass will be marketed as a commercial intermediate product. This hybrid business model enables the project to generate immediate cash flow through black mass sales while simultaneously establishing expertise in downstream metal recovery.

The mechanical recycling section has been designed with state-of-the-art enclosed shredding, crushing, separation and dust collection systems to ensure safe handling of spent lithium-ion batteries while maximizing recovery of black mass, copper, aluminium, steel and plastic fractions. The integration of advanced air pollution control systems, including pulse jet bag filters, cyclone separators and negative pressure ventilation, ensures compliance with the environmental standards prescribed by the Central Pollution Control Board (CPCB) and the concerned State Pollution Control Board.

The hydrometallurgical processing section has been designed to recover valuable metals through controlled leaching, purification, solvent extraction, selective precipitation and crystallization processes. The plant is expected to produce commercially valuable products such as **lithium carbonate, cobalt compounds, nickel compounds, manganese sulphate and recovered graphite**, thereby significantly enhancing value addition compared to the sale of untreated black mass.

From an environmental perspective, the project incorporates robust pollution control infrastructure, including a dedicated **Effluent Treatment Plant (ETP), Zero Liquid Discharge (ZLD)-ready wastewater management system, hazardous waste storage facility**, and dedicated emission control systems for both the mechanical and hydrometallurgical sections. These systems will ensure environmentally sound recycling practices, efficient resource utilization and compliance with all applicable environmental regulations.

The proposed plant layout has been developed considering safe material movement, process segregation, hazardous chemical storage, fire safety, future expansion requirements and operational efficiency. Adequate space has been reserved for future expansion of the hydrometallurgical section, allowing the facility to gradually process the entire quantity of black mass generated in-house without requiring major modifications to the overall plant infrastructure.

Financial evaluation indicates that the project requires an estimated capital investment of approximately **₹11.25 crore**, including land, buildings, plant and machinery, utilities and supporting infrastructure. Based on the projected operating parameters, the facility is expected to generate an annual revenue of approximately **₹33 crore** at full capacity, with a stable operating margin after achieving commercial production. The financial analysis indicates that the project is capable of achieving **operating payback during the fourth year**, demonstrating satisfactory long-term profitability under conservative assumptions without considering additional revenue from EPR Certificates, government subsidies, carbon credits or future expansion.

The project also offers significant opportunities for future value addition. Expansion of the hydrometallurgical section to process the complete black mass generated by the mechanical recycling line can substantially increase profitability. Additional downstream opportunities include the production of battery-grade lithium carbonate, cobalt sulphate, nickel sulphate, manganese sulphate, purified graphite products, copper granules, aluminium granules and engineered recycled plastic products. These value-added products will further improve revenue realization and strengthen the project's competitive position in the critical minerals supply chain.

In addition to its financial attractiveness, the project delivers significant environmental and socio-economic benefits by diverting hazardous lithium-ion batteries from landfills, recovering strategically important critical minerals, reducing greenhouse gas emissions associated with primary mining, conserving natural resources and supporting India's vision of resource security and sustainable manufacturing. The project will also contribute to employment generation, development of recycling infrastructure and promotion of domestic circular economy initiatives.

Considering the increasing generation of end-of-life electric vehicle batteries, consumer electronics batteries and manufacturing scrap, coupled with evolving Extended Producer Responsibility (EPR) obligations and government emphasis on domestic critical mineral recovery, the proposed project possesses substantial long-term growth potential. With appropriate implementation, adherence to statutory compliances, robust operational management and strategic marketing of recovered products, the proposed facility is expected to establish itself as a commercially successful and environmentally responsible lithium-ion battery recycling and critical mineral recovery unit.

Overall, the proposed project demonstrates strong technical feasibility, regulatory compliance, environmental sustainability and financial viability, making it a sound investment opportunity capable of supporting India's transition towards a resilient and self-reliant battery recycling ecosystem while generating sustainable returns for investors over the long term.

22. References

1. Project Data & Internal Inputs

The technical and financial analysis of the project is based on:

- Detailed **plant and machinery specifications** for lithium-ion battery recycling (mechanical processing and hydrometallurgy)
- **CAPEX estimation sheets** for hydrometallurgical processing (2 TPD black mass capacity)
- **OPEX data sheets** including chemical consumption, power, evaporation, and manpower cost
- Process flow and equipment configuration for:
 - Mechanical separation (black mass generation)
 - Hydrometallurgical metal recovery

2. Industry Benchmarks & Technical References

The following industry-standard parameters were used:

- Black mass yield: **50–70% from battery input**
- Metal recovery efficiency:
 - Cobalt/Nickel: **90–95%**
 - Lithium: **85–90%**
- Hydrometallurgical processing cost ranges
- Utility consumption norms (power, water, steam)
- Typical manpower deployment for mid-scale recycling plants

These are based on:

- Global lithium-ion battery recycling practices
- Metallurgical processing benchmarks
- Operational data from existing recycling facilities

3. Market Research References

A. Lithium-Ion Battery Recycling Market (India & Global)

1. **NITI Aayog & Rocky Mountain Institute (RMI)**
 - “*Advanced Battery Recycling in India*”
 - Provides insights on:
 - Battery waste generation
 - Recycling potential
 - Policy framework
2. **International Energy Agency (IEA)**
 - “*Global EV Outlook*”
 - Data on:
 - EV growth
 - Battery demand
 - Future recycling volumes
3. **Deloitte India**
 - “*Lithium-ion Battery Value Chain in India*”
 - Covers:
 - Supply chain gaps
 - Recycling opportunities
4. **McKinsey & Company**
 - “*Battery 2030 / Battery Insights Reports*”
 - Includes:
 - Metal demand projections
 - Recycling economics

B. Industry & Market Intelligence Reports

5. **Research and Markets**
 - *Global Lithium-Ion Battery Recycling Market Report*
6. **Allied Market Research**
 - *Battery Recycling Market Forecast (2022–2030)*
7. **BloombergNEF (BNEF)**
 - EV and battery demand outlook
 - Raw material supply constraints

◆ C. Indian Industry Insights

8. Reports and disclosures from:
 - Indian lithium-ion battery recyclers
 - EV ecosystem stakeholders
 - Battery manufacturers and aggregators

4. Regulatory & Policy References

The project complies with the following regulations:

1. **Battery Waste Management Rules, 2022**
 - Ministry of Environment, Forest and Climate Change (MoEFCC)
2. **Hazardous Waste Management Rules, 2016**
3. **Environment Protection Act, 1986**
4. **Air (Prevention & Control of Pollution) Act, 1981**
5. **Water (Prevention & Control of Pollution) Act, 1974**
6. **Central Pollution Control Board (CPCB) Guidelines**
 - For battery recyclers
 - Hazardous waste handling
 - Emission standards

5. Engineering & Process Design References

- Standard chemical engineering design practices
- Hydrometallurgical processing principles
- Industrial utility design (boilers, cooling systems, ETP)
- Pollution control system design (scrubbers, bag filters, ZLD)

6. Logistics & Cost Benchmarks

- Indian freight and logistics cost benchmarks (₹/km basis)
- Industrial power tariffs and utility costs
- Labour cost benchmarks for industrial operations

7. Financial Modeling References

- Standard project finance methodologies:
 - ROI (Return on Investment)
 - IRR (Internal Rate of Return)
 - Payback Period
- Cost escalation and conservative scenario modeling
- Industry norms for:
 - EBITDA margins
 - Operating cost structures

The project report has been prepared based on a combination of technical data, industry benchmarks, and market research reports. Key references include publications from NITI Aayog, International Energy Agency (IEA), Deloitte, McKinsey, BloombergNEF, and other industry research agencies. Regulatory compliance requirements have been derived from CPCB and MoEFCC guidelines. Financial projections and operational assumptions are based on prevailing industry practices and conservative market estimates.